

IT 320 Course Project Semester-2, 1447H



Software Product Release

<Thameen>

Prepared by

<Amal Albogami>

<Albatool Aljaber>

<Shahd Aldawhan>

<Lina Alsaaran>

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1 Chapter 1: Introduction

In recent years, transportation services have become an essential part of daily life due to the increasing movement of valuable personal and business assets, Items such as cash, jewellery, and high end electronic devices require careful handling and secure transportation methods, especially when transferred between individuals or organizations. As societies continue to grow and rely more on digital services, the transportation sector has evolved to support safer, more organized, and technology driven delivery processes.

The transportation of high-value and sensitive items presents a significant problem due to the lack of standardized, secure, and well coordinated delivery processes, Individuals and organizations often rely on informal or fragmented transportation methods, which increases the risk of loss, damage, or misuse of valuable assets, The absence of clear accountability, real-time tracking, and controlled access further complicates the safe transfer of items such as cash, jewellery, and high-end electronic devices.

Technological solutions play an important role in addressing these challenges at both local and global levels. Locally, within the city of Riyadh, improving the safety and organization of transporting valuable assets can enhance trust among individuals and businesses, reduce security risks, and support a more structured service environment. On a broader level, the development of secure and technologydriven transportation systems aligns with global digital transformation trends and the

increasing demand for reliable handling of sensitive assets. By enhancing transparency, coordination, and accountability, such systems contribute to safer communities and promote higher standards of service quality worldwide.

This chapter provides a general introduction to the domain of high value item transportation and delivery and highlights the challenges associated with handling valuable and sensitive items. It also emphasizes the importance of technological solutions in improving safety and reliability in this area. Additionally, Chapter 1 presents an overview of the project by introducing the problem, the proposed solution, and the product concept, along with the product vision, roadmap, objectives, scope, and the Scrum Team involved in developing the system.

1.1 The Problem

Transporting high-value and sensitive items is a common challenge in the transportation. Items such as money, jewelry, and expensive electronic devices require a higher level of security compared to regular deliveries. However, most available delivery services are designed for general-purpose transportation and do not fully consider the specific requirements of handling valuable goods. This creates a problem for individuals and businesses that need safe and reliable ways to transport such items.

For instance, a jewelry store that needs to deliver valuable products to customers cannot depend on standard delivery applications. These services usually do not provide secure storage, tracking, or proper accountability during transportation. As a result, valuable items may be exposed to risks such as theft, loss, or damage, especially when the delivery process is informal or not well monitored.

This project narrows its focus to a specific aspect of the issue, which is the limited availability of secure and trustworthy transportation vehicles dedicated to carrying valuable items. While the risks of transporting valuable goods are widely recognized, individuals and businesses often face difficulty in obtaining a reliable vehicle that can be trusted to handle such items safely. The lack of accessible and dependable transportation options creates uncertainty and discourages the safe movement of valuable goods.

1.2 The Solution

To address this problem, a specialized web-based transportation system is proposed to support the secure delivery of items that require a high level of protection, including jewelry, money, and expensive electronic devices. The system relies on dedicated vehicles equipped with secure safes to ensure item protection throughout the transportation process. To further enhance security, access to transported items is controlled through a confidential security code known only to the sender and the receiver.

In addition, the system provides insurance coverage for transported items, offering compensation in case of loss or damage. By combining secure transportation methods with a structured and reliable delivery process, the proposed solution aims to provide individuals and shops with a trustworthy and safe alternative to traditional delivery services.

1.3 The Product

1.3.1 Product Vision

In this section we will describes the product vision of Thameen

Product Vision:

For individuals and local businesses within Riyadh

Who need a secure and reliable method to transport high value items within the city

The Thameen is a web based transportation management system

That aims to provide a safer and more controlled way to transport high value items.

Unlike existing secure transportation providers such as **APSG World**, which mainly rely on traditional delivery methods

Our product enables safe transportation of valuable items using special vehicles with secure lockers and one time access codes created by the customer

1.3.2 Product Roadmap

In this section we will show the Product Roadmap of Thameen



Figure 1 product roadmap

1.3.3 Objectives

In this section, we define the product, project, and learning objectives related to the development of “Thameen” web-based transportation system. These objectives represent the goals that the project aims to achieve by the end of the development process.

Product Objectives (Customer focus – value) :

These objectives describe how the Thameen system addresses the problem of securely transporting high-value items and provides value to individuals and shops.

- Enable users to register in the system.
- Enable users to log in the system.
- Enable users to log out of the system.
- Allow users to add new transportation requests.
- Enable users to search for their requests using a unique transportation request number.
- Enable users to view their request details and status.
- Allow users to edit their request details before final submission.
- Allow users to delete their request before delivery is finished.
- Allow users to rate the service after successful delivery.
- Allow users to submit a written review after successful delivery.
- Enable administrators to add driver accounts.
- Enable administrators to delete driver accounts.
- Provide an administrative dashboard page with weekly summary reports of delivery requests by service category along with customer ratings and reviews.

Provide insurance coverage for all deliveries to reduce risks related to loss or damage.

Project Objectives (Solution focus – plan) :

These objectives describe the main steps and activities required to successfully complete the development of the Thameen system.

- Understand the problem domain and study secure transportation services related to highvalue items.

- Identify and understand user needs by analyzing the requirements of individuals and shops.
- Define and document system requirements based on user needs and project goals.
- Design the system interfaces and overall structure for different system roles.
- Design and implement the database required to store system data.
- Develop and implement the main functionalities of the system.
- Test the system to ensure it works correctly, securely, and meets user requirements.

Learning Objectives (Student focus) :

These objectives focus on the new technical and practical skills that the team aims to gain through completing the Thameen project, beyond the topics covered in the course curriculum.

- Gain experience in developing a complete web-based system from initial design to final implementation using new programming tools and technologies.
- Learn how to work with web development frameworks and libraries to build structured and scalable real-world applications.
- Develop skills in designing and implementing databases that support secure data storage, reporting, and system management.
- Gain practical experience in applying secure design practices when developing systems that handle sensitive and high-value items.
- Learn how to design simple administrative dashboards and generate basic analytical reports to support system monitoring and decision-making.

1.3.4 Scope

Thameen is developed to support individuals and shops who require a secure and organized transportation service. It is supported on Google Chrome, Microsoft Edge, and Safari. The scope of this project is defined through platform support, language support, and geographic coverage.

The service is provided only within Riyadh city and does not serve any other regions or cities. The system supports the English language. Thameen's service is limited to transportation and delivery only, meaning it does not provide any additional services beyond delivering shipments from one point to another.

Thameen focuses specifically on delivering valuable items, including jewelry, money, and expensive electronic devices. The service does not include transporting living creatures, weapons,

or archaeological/antique artifacts. In addition, Thameen's responsibility and service coverage end immediately once the customer signs to confirm receiving the shipment.

Advanced or extended features such as real-time GPS tracking, online payment integration, or expansion to cities outside Riyadh are considered outside the scope.

1.4 The Scrum Team

Table 1: scrum team table

Scrum Team	
Product Owner:	Lina Alsaaran
Developers:	Amal Albogami Albatool Aljaber Shahd Aldawhan Lina Alsaaran
Scrum Master:	Ghaida Alfayez
Stakeholders:	Ghaida Alfayez

2 Chapter 2: Domain Analysis

This chapter analyzes existing delivery platforms used for transporting valuable items, It reviews websites such as TNT and Breakwells to understand their features, strengths, and limitations, The chapter also includes a comparison between these systems to identify missing or limited features, Finally, the chapter explains how Thameen differs from these existing systems.

2.1 Literature Review

TNT:

[3]

A global courier and logistics company website that represents an international shipping provider offering delivery services for both valuable and regular goods through an online service platform.

Strengths:

- 1- The system provides a shipping cost calculator based on volumetric weight.
- 2-The system supports selecting different transportation modes for shipment delivery (air, road, and sea).
- 3-The system provides a search bar that allows users to enter a shipment number to retrieve shipment information.

Limitations:

- 1- Services are mainly designed for businesses rather than individual users.
- 2- The system does not support local short-distance delivery within the same city.

Breakwells:

[4]

A transport and haulage company website that presents a regional logistics provider specializing in moving different types of goods, including fragile and high-value items, and providing information about its transportation services.

Strengths:

- 1- The system provides a blog section that displays logistics and shipping articles to help users understand shipping procedures and requirements.

2- The system integrates and displays customer reviews from Google directly on the website to allow users to evaluate service quality.

3- The system categorizes shipments into multiple item types such as fragile goods, heavy items, furniture, and hospital equipment.

Limitations

1-The navigation menu is not immediately noticeable, which may confuse first-time users when trying to find the list of services.

Table 2:feature comparsion between competitors systems

feature	TNT	breakwells	thameen
Select the delivery vehicle type	yes	no	no
View reservation details	yes	yes	yes
Service rating	no	yes	yes
select the preferred delivery time	no	yes	no
Providing a locked safe box secured with a code for transporting valuable items	no	no	yes

Although several existing systems specialize in transporting valuable items, they mainly depend on general vehicle security without ensuring controlled access to the shipped item itself. In many cases, the user must rely only on trust in the driver or company, which may create concerns when sending highly valuable belongings.

Our proposed system improves the level of protection by introducing a secure safe box installed inside the delivery vehicle. The safe box can only be opened using a unique access code shared

exclusively between the sender and the intended receiver, preventing unauthorized access during transportation. Moreover, the system provides insurance coverage for shipments to protect users against potential loss or damage.

In addition, users are able to monitor the shipment status and verify the delivery process, ensuring that the item is handed over to the correct person. By combining protected storage, controlled access, shipment insurance, and delivery verification, the proposed system provides higher reliability and trust compared to existing competing solutions.

3 Chapter 3: Requirements Engineering

This chapter describes the requirements engineering activities for Thameen system, To ensure that system requirements are clearly identified, analyzed, and documented, The main goal of this chapter is to capture users needs and translate them into clear and structured system requirements that support the secure transportation of high value items.

The chapter begins with requirements elicitation and analysis, where user interviews and questionnaires were used to gather requirements from the target users, Interviews helped understand users concerns, expectations, and previous experiences related to transporting valuable items, while questionnaires were used to collect feedback from a wider group of users The results and findings from both methods were analyzed to identify the key functional and non-functional requirements of the system.

Next, the System Users section presents a detailed description of the different user types involved in the system, including their characteristics, roles, and level of interaction with the system This section helps clarify user responsibilities and ensures that the system features are designed to match user needs.

Then a Use Case Diagram is presented to visually represent the interactions between the system and its users, showing the main functionalities supported by Thameen system, Finally, this chapter concludes with the Product Backlog, which organizes the identified functional and non-functional requirements into user stories, defining the scope, priorities, and acceptance criteria for system development.

3.1 Requirements Elicitation and Analysis

Two requirement elicitation methods were used in this project to identify and analyze the system requirements for the Thameen transportation system, Interviews and questionnaires were conducted with individuals and shop owners to understand their needs, concerns, and expectations regarding the secure transportation of high value items, Using these two methods helped ensure that the collected requirements were accurate, relevant, and aligned with real user needs.

3.1.1 Interviews

Four interviews were conducted with participants from the target user groups, including individuals and shop owners, Each interview consisted of questions aimed to understand users experiences and preferences related to transporting high value items, The interview questions and full transcriptions are provided in Appendix A.

Participants Information:

Table 3: participants information

Name	Age	Gender	Nationality	User Type
Norah	38	Female	saudi	Shop owner
Faisal	28	Male	saudi	Shop owner
Reem	21	Female	saudi	individual
Rawan	24	Female	saudi	individual

Results and Findings:

The interviews revealed several important findings related to how users perceive transporting valuable items and what they expect from a secure transportation system, Although the participants had different backgrounds and usage patterns, common concerns and expectations appeared across the interviews, The interview questions and full transcriptions are included in Appendix A

One key finding is that all participants had previous experience transporting valuable items, but the frequency and context differed, The shop owners (Interview 1 and Interview 2) described transporting valuable items as a regular part of their business operations, which made reliability and consistency very important to them, On the other hand , individual users (Interview 3 and Interview 4)

said that they usually transport valuable items on specific occasions, such as sending personal items or valuables to family members, which often caused feelings of anxiety and concern.

Security and proper handling were major concerns mentioned by all participants, Shop owners emphasized the risk of mistakes, loss, or damage during delivery, as these issues directly affect customer trust and business reputation (Interview 1, Interview 2), Individual users focused more on personal worries, such as not knowing who is handling the item and whether it is being treated carefully (Interview 3), One participant specifically highlighted privacy as a concern, worrying about unnecessary access to personal shipment information (Interview 4).

Another important finding relates to trust and transparency in the delivery process, Participants expressed that feeling informed about how items are handled increases their sense of comfort, For example, (Interview 1) highlighted the importance of clear handling procedures and controlled access, while (Interview 4) mentioned that an organized and professional process from start to finish builds trust in the service, This shows that trust is influenced not only by security measures but also by how clear and structured the service appears to users.

Regarding service interaction, all participants preferred using a digital system to request transportation services. However, their expectations differed slightly, Shop owners preferred efficient and structured systems that support frequent use and detailed submissions (Interview 2), while individual users preferred simple and easy to use applications that do not require many steps (Interview 4), This indicates that ease of use is important, especially for users who do not use delivery services regularly.

Finally, All participants expressed a preference for specialized transportation services over general delivery services, Shop owners preferred specialized services because they focus on protecting valuable goods and maintaining reliability for business needs (Interview 1, Interview 2), Individual users preferred specialized services when they felt trustworthy and simple, even if they used them only occasionally (Interview 4), This suggests that specialization increases confidence across different users type.

Overall, the findings show that while users have different experiences and needs, security, trust, clear processes, and appropriate service design play a crucial role in shaping their expectations of valuable item transportation services.

3.1.2 Questionnaire

A questionnaire was conducted to gather quantitative feedback from potential users regarding the proposed Thameen system. The questionnaire consisted of six questions focusing on insurance coverage, delivery tracking, deletion policy, rating functionality, item type, and pricing preferences.

The questionnaire was distributed online, and a total of 11 participants responded.

Analysis of Results:

For the first question, 54.5% of participants indicated that they would most likely transport jewelry, followed by 27.3% for cash and 18.2% for electronics. This supports categorizing deliveries based on item type within the system, allowing better organization and service management.

Regarding insurance coverage, the majority of participants confirmed that insurance would increase their trust in the service, reinforcing its inclusion as a core feature.

Most participants considered delivery tracking to be very important, supporting the implementation of real-time status updates (Pending, In Transit, Delivered).

Concerning the deletion policy, the majority found it reasonable to allow users to delete a request before it becomes “Delivered,” supporting the system’s controlled flexibility.

In addition, participants indicated that rating functionality after delivery is important, justifying the inclusion of customer feedback features.

Finally, most respondents preferred fixed pricing for all deliveries rather than pricing based on item type and value, supporting the adoption of a fixed pricing strategy.

Summary of Questionnaire Findings:

Overall, the questionnaire results support the inclusion of item categorization, insurance coverage, real-time delivery tracking, controlled deletion policy, rating functionality, and a fixed pricing strategy within the Thameen system. Participants showed strong interest in secure transportation of valuable items, emphasized the importance of insurance and tracking features, and preferred fixed pricing for deliveries. These findings validate the core functional requirements proposed for the system.

Summary of Requirements Findings:

Based on the interviews and questionnaire analysis, several key system requirements were identified, Users showed the importance of secure transportation for valuable items, highlighting concerns about safety, privacy, and proper handling, Trust and transparency were also critical, as users prefer clear procedures and visibility throughout the delivery process.

The findings showed that both shop owners and individual users require a reliable and easy to use system, although their needs differ slightly, Shop owners prefer structured systems that support frequent and detailed submissions, while individual users prefer simple interfaces with minimal steps.

Additionally, participants strongly supported features such as item categorization, insurance coverage, real-time tracking (Pending, In Transit, Delivered), and a controlled deletion policy before delivery completion, Rating functionality after delivery was also considered important to enhance feedback and service quality.

Finally, most participants preferred fixed pricing over variable pricing, as it simplifies the process and increases user confidence, Overall, the analysis confirms that the system should focus on security, usability, transparency, and trust to meet user expectations effectively.

3.2 System Users

Thameen has two main user types: customers (individuals and shops) who request secure transportation, and administrators who manage the system. Users are expected to be adults and live in Riyadh. Since the system is in English, users should be able to read basic English and have basic technical skills to use a web-based service.

- Customers: Adults (18+), Saudi or non-Saudi residents in Riyadh. No specific education level is required. They should have basic familiarity with online services to submit requests and follow order updates.
- Administrators: Adults (21+). Admin users have good technical expertise and system knowledge. They monitor and manage orders and users, and they also supervise drivers by reviewing their performance, handling complaints, and removing drivers if needed.

3.3 Use Case Diagram

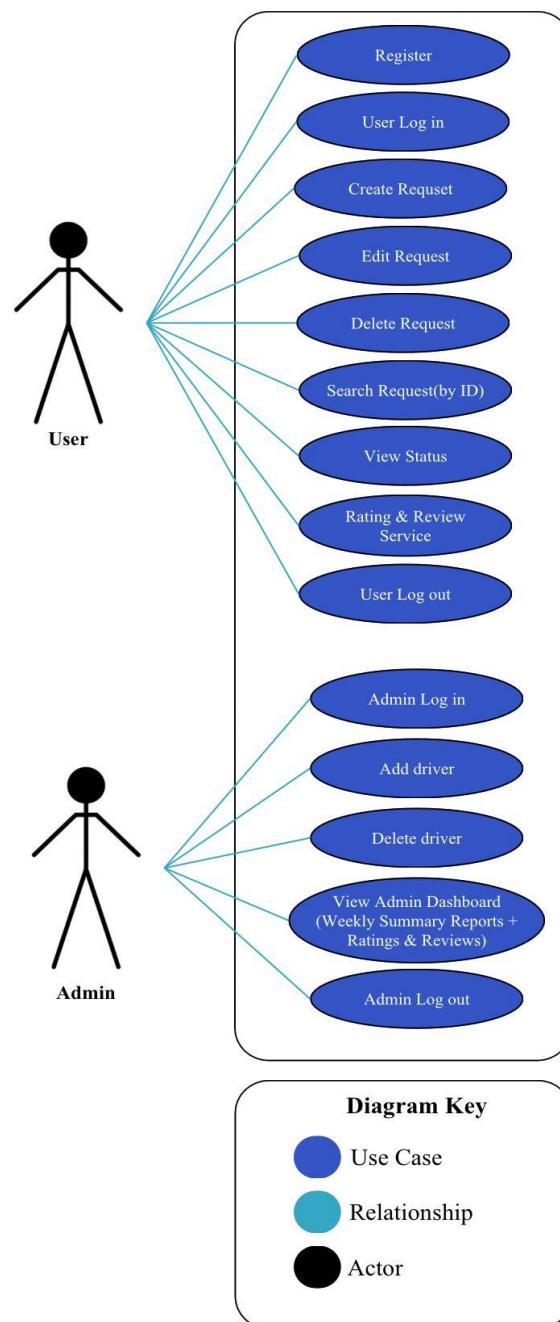


Figure 2: use case diagram

3.4 Product Backlog

Table 4: product backlog

PBI	Size	Type	Acceptance Criteria
As a user I want to register so that I can access Thameen services and create delivery requests.	3	Feature	- If I enter a valid Full Name, Phone Number, Date of Birth shows I am 18 years or older, and a Password with a minimum of 8 digits and click Register, then the system should create my account successfully and allow me to access the system by redirecting me to Login. - If I enter a Date of Birth that indicates I am under 18 years old, and click Register, then the system should display an error message stating that I must be at least 18 years old to register. - If I enter a Phone Number that already exists and click Register, then the system should reject the registration and display an error message indicating the phone number is already registered. - If I leave Full Name, Phone Number, or Date of Birth empty, or enter a Password with less than 8 digits, and click Register, then the system should display a validation error and prevent account creation until the inputs are valid.
As a user I want to log in using my phone number and password so that I can access Thameen services and manage my delivery requests.	2	Feature	- If I go to the customer login page and enter a registered Phone Number and the correct Password and click "Login", then the system should authenticate me successfully and allow me to access the system. - If I go to the customer login page and enter an unregistered Phone Number or an incorrect Password and click Login, then the system should deny access and display a generic error message stating that the Phone Number or password is incorrect, without specifying which one for security reasons.

			• If I leave the Phone Number or Password empty and click Login, then the system should display a validation error and prevent login until the inputs are provided.
As an admin, I want to log in to the system so that I access administrative functions.	2	Feature	• If I go to the admin login page and enter a correct username and password, the system allows access to the admin dashboard. • If I go to the admin login page and enter incorrect username or password, then the system displays a generic error message stating that the login credentials are invalid, without specifying which one is wrong for security reasons. • Only authorized administrators can access the admin interface.
As a user, I want to log out from my account so that I can protect my personal and shipment information.	2	Feature	• When the user clicks the “Logout” button, the system ends the session. • The user is redirected to the login page. • The user cannot access account pages after logout unless logging in again.
As an admin, I want to log out from the admin dashboard so that I can protect system data and shipment management information.	2	Feature	• When the admin clicks the “Logout” button, the system ends the admin session. • The admin is redirected to the admin login page. • The admin cannot access dashboard or management pages after logout unless logging in again.
As a user I want to add a delivery request so that I can transport valuable items securely within Riyadh.	5	Feature	• If I am logged in, select an allowed item type (Jewelry, Cash, or High-value Electronics), choose an item value range (Less than 5,000 SAR, 5,000-10,000 SAR, or More than 10,000), enter a pickup location and drop-off location within Riyadh, enter a 4-digit security code that will be used to open the secure locker, and confirm it, and click “Create Request”, then the system should

			create the request successfully and display a confirmation with a unique Order Number, and show the service price and insurance coverage (up to 10,000 SAR) before final submission. • If I leave any required field (item type, item value range, pickup location, or drop-off location) empty and click "Add Request", then the system should display a validation error and prevent request creation until all required inputs are provided. • If I enter a pickup or drop-off location outside Riyadh, then the system should reject the request and display an error message explaining the restriction. • If I select an item value range above 10,000 SAR, then the system should display a message informing me that the insurance coverage included in the service is limited to 10,000 SAR. • The system should automatically calculate and display the service price based on the selected item type: · Jewelry: 200 SAR · Cash: 150 SAR · High-value Electronics: 120 SAR The price must be shown before final submission, and if I confirm, the request is created with the calculated price. • The security code is stored securely and is never displayed to anyone, including the driver. Only the sender and the receiver know the code. Upon delivery, the receiver must provide this code to the driver to unlock the locker and complete the delivery.
As a user, I want to view my request details and delivery status so	2	Feature	• If I log in to my account and open my requests page, then I should be able to view the details of my request.

that I see the progress of my shipment.			- The system should display the current delivery status of the request (Pending, In Transit, or Delivered). - If I open a specific request, then the system should show the item type, item value range, pickup location, drop-off location, and order number. - The request details and status should be shown in a clear and readable format. - If I open my requests page and have no delivery requests, then the system should display a message indicating that no requests are available. - Only the owner of the request should be able to view its details and status.
As a user, I want to search for my request using a request ID so that I can quickly find a specific request.	3	Feature	- If I enter a valid request ID in the search field, then the system should display the corresponding request details. - If the entered request ID does not exist, then the system should display an error message. - The search results should only show requests that belong to the logged in user.
As a user I want to edit my delivery request before final submission so that I can correct or update the request details.	3	Feature	- If my delivery request is not finally submitted and I click Edit, then the system should display the current delivery request details, including the item type, item value range, and the pickup and drop-off locations within Riyadh. - If the modified details (item type, item value range, pickup location, or drop-off location) and I click “Save”, and the pickup and drop-off are within Riyadh, then the system should update the delivery request with the new information and display a success message. - If I select an item value range above 10,000 SAR, then the system should display a message informing me that

			the included insurance covers shipments up to 10,000 SAR. • If I leave any required field empty and click “Save”, then the system should display a validation error and prevent the update until all required fields are filled. • If the delivery request is already finally submitted, then the system should not display the “Edit” option for that request, and the user cannot modify it.
As a user, I want to delete my delivery request before it is marked as “Delivered” so that I can cancel it if I no longer need the service.	2	Feature	• If I am logged in and viewing a delivery request that is not yet marked as “Delivered”, and I click “Delete”, then the system should ask for confirmation (e.g., “Are you sure you want to delete this request? This action cannot be undone. “). • If I confirm the deletion, the system should permanently remove the request from my list and display a success message (e.g., “Request deleted successfully. “). • If I cancel the confirmation, the system should close the confirmation dialog and keep the request unchanged. • If the delivery request is already marked as “Delivered”, the system should not display the “Delete” option for that request, and the user cannot delete it.
As an admin, I want to add driver accounts so that I can assign delivery requests to available drivers.	3	Feature	• If I enter a valid driver full name, phone number, national ID, and vehicle details and click “Add Driver”, then the system should create the driver account successfully. • If any required field is empty, then the system should display a validation error message and prevent submission. • If I enter a phone number or national ID that already exists, then the system should reject the request and display an error message.

			- The newly added driver should appear in the drivers list. - Only logged-in administrators can add drivers.
As an admin, I want to delete a driver account so that I can remove inactive or incorrect driver records.	2	Feature	- If I select a driver and click “Delete”, the system should show a confirmation message. - If I confirm, the driver should be removed from the system. - After deletion, the driver should no longer appear in the list.
As a user, I want to rate the delivery service after successful delivery so that I can share my satisfaction level.	3	Feature	- If the delivery status is “Delivered”, the user can select a rating from 1 to 5 stars. - The system should save the rating successfully. - The rating should be included in the weekly average rating displayed in the admin dashboard. - A confirmation message should be displayed after submission.
As a user, I want to submit a written review after the delivery is marked as Delivered so that I can provide detailed feedback to the admin.	3	Feature	- If the delivery status is “Delivered”, the user can submit a written review. - The review text field must not be empty. - The system should save the review successfully. - The review should be displayed in the admin dashboard under the reviews section. - A confirmation message should be displayed after submission.
As an admin, I want to view a dashboard that shows a weekly summary of delivery requests so that I can monitor the system.	3	Feature	- The dashboard page is accessible only to loggedin admins. - The dashboard displays the current week summary automatically. - The dashboard shows:

			○ Total number of delivery requests created this week ○ Number of requests in each status (Pending / In Transit / Delivered) • The dashboard displays all ratings and written reviews submitted this week. • The dashboard shows the average rating for this week.
As a user, I want my delivery request to include insurance coverage so that my valuable items are protected during transportation.	3	Feature	• The system should automatically include insurance coverage up to 10,000 SAR for every delivery request. • The insurance coverage amount should be fixed. • The insurance information should be saved with the request. • The insurance coverage information should be displayed within the request details.
As a user, I want the system pages to load quickly so that I can track and create delivery requests without delay.	2	Feature	• Main pages load within 5 seconds. • Shipment status page loads within 5 seconds after request. • The system handles multiple users without crashing.
As a user, I want the system to be available whenever I access it so that I can check shipment status at any time.	2	Feature	• The website is available 24/7. • If the server fails, the system restores service quickly. • Users can access shipment tracking at any time.

4 Chapter 4: System Design

This chapter presents the overall design of the proposed system, focusing on how its components are structured and how they interact to deliver the required functionality. It provides a comprehensive view of the system by illustrating its architecture, data structure, and component organization.

The purpose of this chapter is to transform the system requirements into a clear and organized design that supports scalability, maintainability, and efficient performance. It includes several design models such as system architecture, class diagrams, data design, and interface design.

Through these models, the chapter demonstrates how the system is divided into different components and how responsibilities are distributed among them to ensure a reliable and user-friendly solution.

4.1 System Architecture

The architecture selected for Thameen is the Client–Server architecture. This architecture was chosen because it provides a clear separation between the user interface, the application logic, and the database. Such separation makes the system easier to develop, maintain, secure, and scale. It is also suitable for Thameen because the system depends on continuous communication between the client side, where users and administrators interact with the application, and the server side, where requests are processed and data is managed.

At a high level, the system consists of three main components: the client, the server, and the database. The client side represents the application interface used by both users and administrators. Through this interface, users can register, log in, create delivery requests, view requests, edit or delete their requests, rate and review deliveries, and log out. Administrators can log in and perform administrative operations such as adding or deleting drivers and viewing the admin dashboard. These actions are sent through the Internet to the server in the form of requests.

The server is responsible for the core system functionality. It handles authentication and validation, processes incoming requests, applies the business logic, retrieves data from the database, and performs operations such as adding, updating, and deleting records. The server communicates directly with the system database, ThameenDB, which stores all essential system data, including user information, delivery requests, driver data, ratings, and other related records. After processing each request, the server returns an appropriate response to the client.

This architecture was selected because it ensures organized distribution of responsibilities among system components. The client focuses on presentation and user interaction, the server manages processing and decision-making, and the database ensures reliable data storage and retrieval. As a result, the Client–Server architecture provides a reliable, maintainable, and user-friendly solution for the Thameen system.

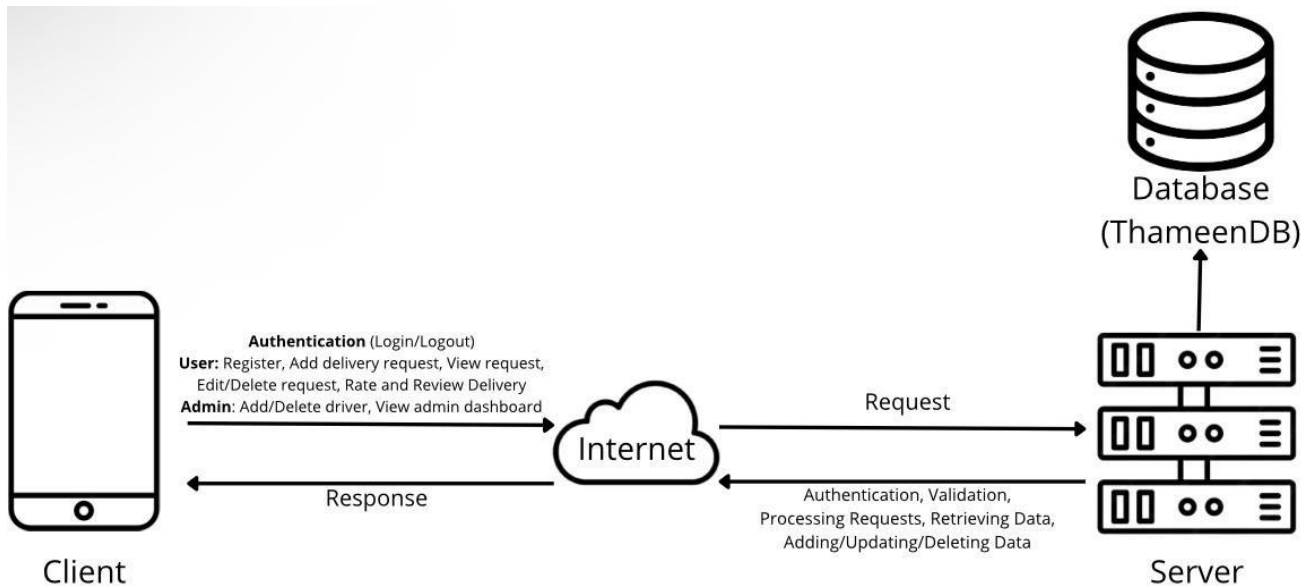


Figure 3: system architecture

4.2 Class Diagram

In this section, we will represent the class diagram of " Thameen " app. We provide a detailed representation of the classes in our system, representing their attributes, methods, and relationships. The Class Diagram serves as a visual plan that shows the structure and organization of the code developed during implementation of the object-oriented design of " Thameen ".

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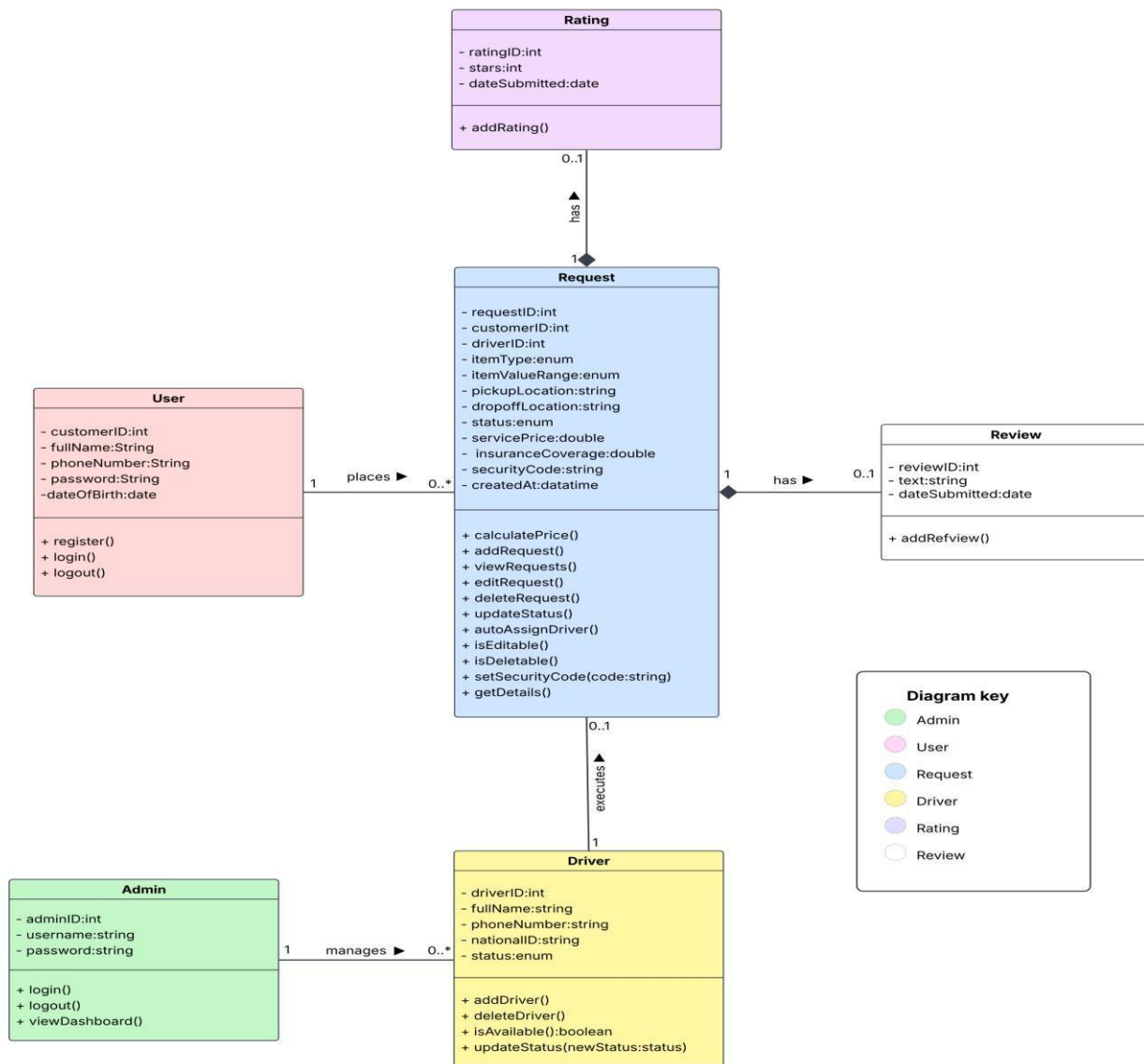


Figure 4: class diagram

4.3 Data Design

This section presents the data design of Thameen system, Since the system uses a SQL database, the data design includes the ER diagram, the relational schema, and the data dictionary. These elements describe the main entities of the system, their attributes, and the relationships between them in a structured relational format.

ER Diagram:

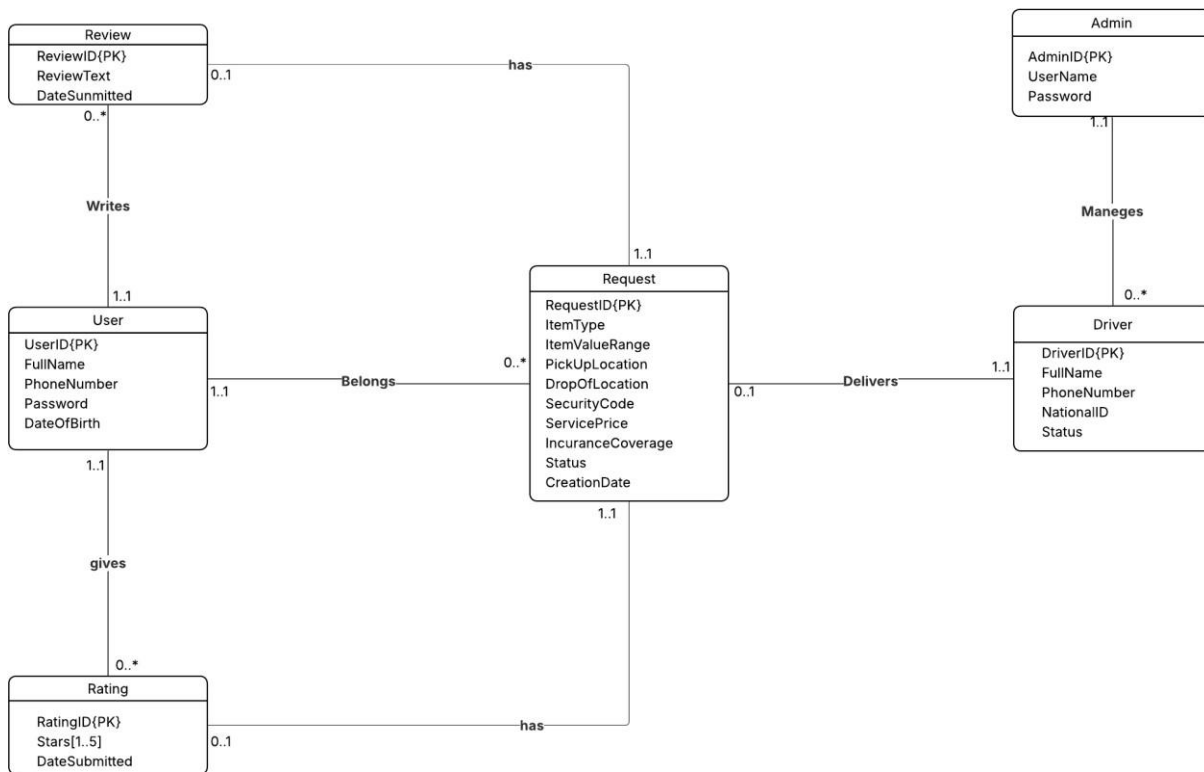


Figure 5: ER digram

Relational Schema:

User (UserID, FullName, PhoneNumber, Password, DateOfBirth) Primary

Key: UserID

Admin (AdminID, UserName, Password)

Primary Key: AdminID

Driver (DriverID, FullName, PhoneNumber, NationalID, Status, AdminID)

Primary Key: DriverID

Foreign Key: AdminID references Admin(AdminID)

Request (RequestID, ItemType, ItemValueRange, PickupLocation, DropOffLocation, SecurityCode, ServicePrice, InsuranceCoverage, Status, CreationDate, UserID, DriverID)

Primary Key: RequestID

Foreign Key: UserID references User(UserID)

Foreign Key: DriverID references Driver(DriverID)

Rating (RatingID, Stars, DateSubmitted, UserID, RequestID)

Primary Key: RatingID

Foreign Key: UserID references User(UserID)

Foreign Key: RequestID references Request(RequestID)

Review (ReviewID, ReviewText, DateSubmitted, UserID, RequestID)

Primary Key: ReviewID

Foreign Key: UserID references User(UserID)

Foreign Key: RequestID references Request(RequestID)

Data Dictionary for Entities:

Entity Name	Description	Aliases	Occurrence
User	A user is a customer who creates delivery requests to transport valuable items through the system.	Customer	A user can create zero or many requests. A user can also submit ratings and reviews for completed requests.
Admin	An admin is responsible for managing the system, viewing ratings and reviews, and handling drivers.	-	An admin can manage zero or many drivers and access the admin dashboard.
Driver	A driver is responsible for transporting valuable items from the pickup location to the dropoff location.	-	A driver can deliver zero or one request. Each driver is managed by one admin.
Request	A request represents a delivery order created by a user for transporting valuable items.	-	Each request belongs to one user and is assigned to one driver. A request can have zero or one rating and zero or one review.
Rating	A rating represents the user's numerical evaluation of a completed delivery request.	-	A rating belongs to one request and is submitted by one user.
Review	A review represents the user's written feedback about a completed delivery request.	-	A review belongs to one request and is written by one user.

Table 5: data dictionary for entites

Data Dictionary for Relationships:

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
User	1..1	Creates	Request	0..*
Admin	1..1	Manages	Driver	0..*

Driver	1..1	Delivers	Request	0..1
User	1..1	Gives	Rating	0..*
Request	1..1	Has	Rating	0..1
User	1..1	Writes	Review	0..*
Request	1..1	Has	Review	0..1

Table 6: data dictionary for relationships

Data Dictionary for Attributes:

Entity Name	Attribute	Description	Data Type	Length	Nulls	Multi-Value	Default Value	Range	PK
User	UserID	Unique identifier for the user	Integer	-	N	N	-	-	Y
User	FullName	User full name	String	50	N	N	-	-	N
User	PhoneNumber	User phone number	String	10	N	N	-	-	N
User	Password	User account password	String	20	N	N	-	-	N

User	DateOfBirth	User date of birth	Date	-	N	N	-	-	N
Admin	AdminID	Unique identifier for the admin	Integer	-	N	N	-	-	Y

Admin	UserName	Admin username	String	50	N	N	-	-	N
Admin	Password	Admin account password	String	20	N	N	-	-	N
Driver	DriverID	Unique identifier for the driver	Integer	-	N	N	-	-	Y
Driver	FullName	Driver full name	String	50	N	N	-	-	N
Driver	PhoneNumber	Driver phone number	String	10	N	N	-	-	N
Driver	NationalID	Driver national identification number	String	10	N	N	-	-	N
Driver	Status	Current availability status of the driver	Enum	-	N	N	Available	Available / Busy	N
Request	RequestID	Unique identifier for the request	Integer	-	N	N	-	-	Y

Request	ItemType	Type of valuable item to be delivered	Enum	-	N	N	-	Jewelry / Cash / High value Electronics	N
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Request	ItemValueRange	Value range of the item	Enum	-	N	N	-	Less than 5,000 / 5,000–10,000 / More than 10,000	N
Request	PickUpLocation	Pickup location of the request	String	150	N	N	-	Riyadh locations only	N
Request	DropOffLocation	Drop-off location of the request	String	150	N	N	-	Riyadh locations only	N
Request	SecurityCode	Secret code used for delivery verification	String	4	N	N	-	-	N
Request	ServicePrice	Calculated delivery service price	Double	-	N	N	-	120–200 SAR	N
Request	InsuranceCoverage	Insurance amount included in the request	Double	-	N	N	10000	Up to 10,000 SAR	N
Request	Status	Current request status	Enum	-	N	N	Pending	Pending / In Transit / Delivered	N
Request	CreationDate	Date and time when the request was created	DateTime	-	N	N	-	-	N

Rating	RatingID	Unique identifier for the rating	Integer	-	N	N	-	-	Y
Rating	Stars	Numerical rating value given by the user	Integer	-	N	N	-	1-5	N
Rating	DateSubmitted	Date when the rating was submitted	Date	-	N	N	-	-	N
Review	ReviewID	Unique identifier for the review	Integer	-	N	N	-	-	Y
Review	ReviewText	Written feedback submitted by the user	String	255	N	N	-	-	N
Review	DateSubmitted	Date when the review was submitted	Date	-	N	N	-	-	N

Table 7: data dictionary for attributes

4.4 Components Design

Create request Flow chart:

ADD REQUEST FLOWCHART

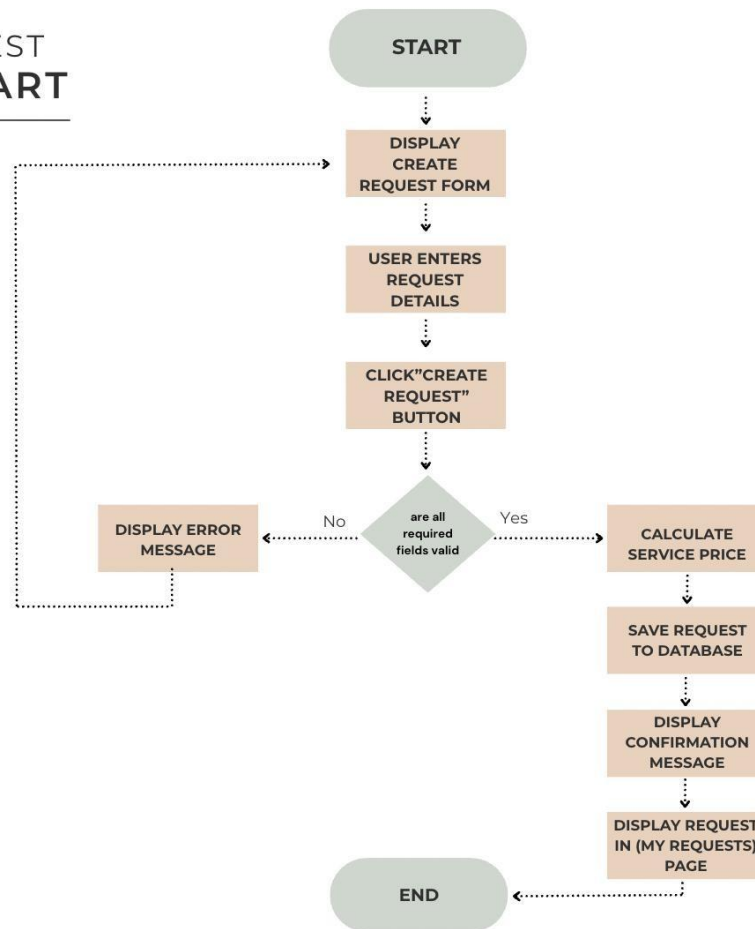


Figure 6: add request flowchart

Pseudocode:

BEGIN

IF user opens "Create Request" page:

 DISPLAY create request form

 DISPLAY "Create Request" button

 SET isValid = false

 WHILE isValid == false:

 IF user enters request details:

 IF user clicks on "Create Request" button:

 VALIDATE request information

IF all required fields are valid:

SET isValid = true

CALCULATE service price

SAVE request to database

DISPLAY confirmation message

DISPLAY request in "My Requests" page

ELSE:

DISPLAY error message

RETURN to create request form

ENDIF

ENDIF

ENDIF

ENDWHILE

ENDIF

END

Edit request flowchart:

EDIT REQUEST FLOWCHART

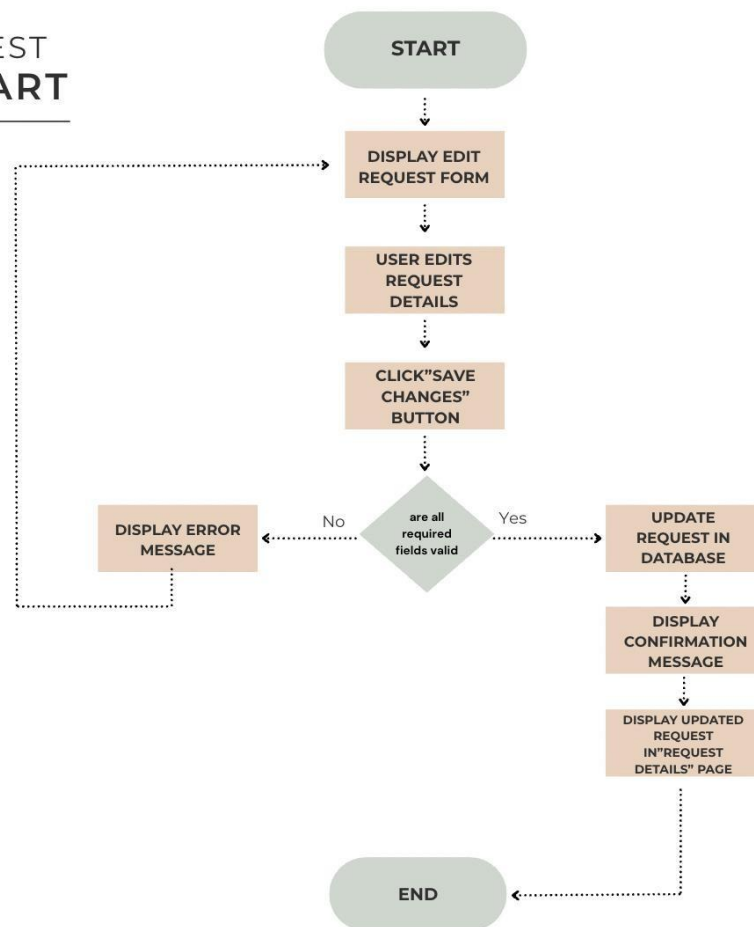


Figure 7: edit request flowchart

Pseudocode:

BEGIN

IF user opens "Edit Request" page:

 DISPLAY edit request form

 DISPLAY "Save Changes" button

 SET isValid = false

 WHILE isValid == false:

 IF user edits request details:

 IF user clicks on "Save Changes" button:

 VALIDATE updated request information

IF all updated fields are valid:

SET isValid = true

UPDATE request in database

DISPLAY confirmation message

DISPLAY updated request in "My Requests" page

ELSE:

DISPLAY error message

RETURN to edit request form

ENDIF

ENDIF

ENDIF

ENDWHILE

ENDIF

END

Delete request flowchart:

DELETE REQUEST FLOWCHART

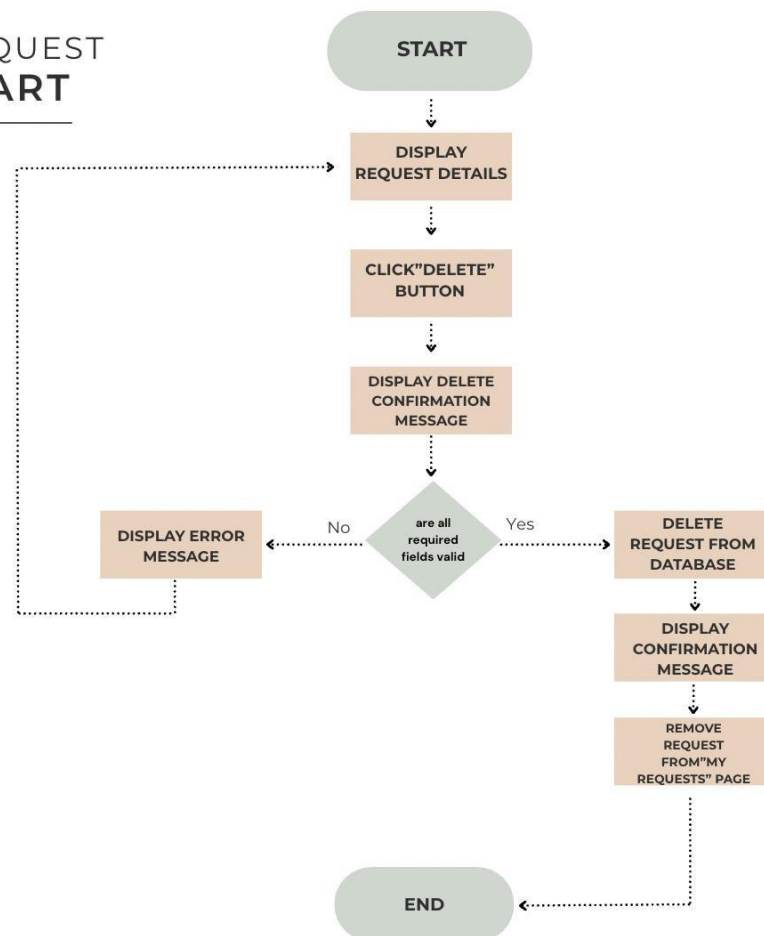


Figure 8: delete request flowchart

Pseudocode:

Pseudocode:

BEGIN

IF user opens "Request Details" page:

 DISPLAY request details

 DISPLAY "Delete" button

 SET isDeleted = false

 WHILE isDeleted == false:

 IF user clicks on "Delete" button:

DISPLAY delete confirmation message

IF user confirms deletion:

SET isDeleted = true

DELETE request from database

DISPLAY confirmation message

REMOVE request from "My Requests" page

ELSE:

DISPLAY error message

RETURN to request details page

ENDIF

ENDIF

ENDWHILE

ENDIF

END

4.5 Interface Design

Sitemap:

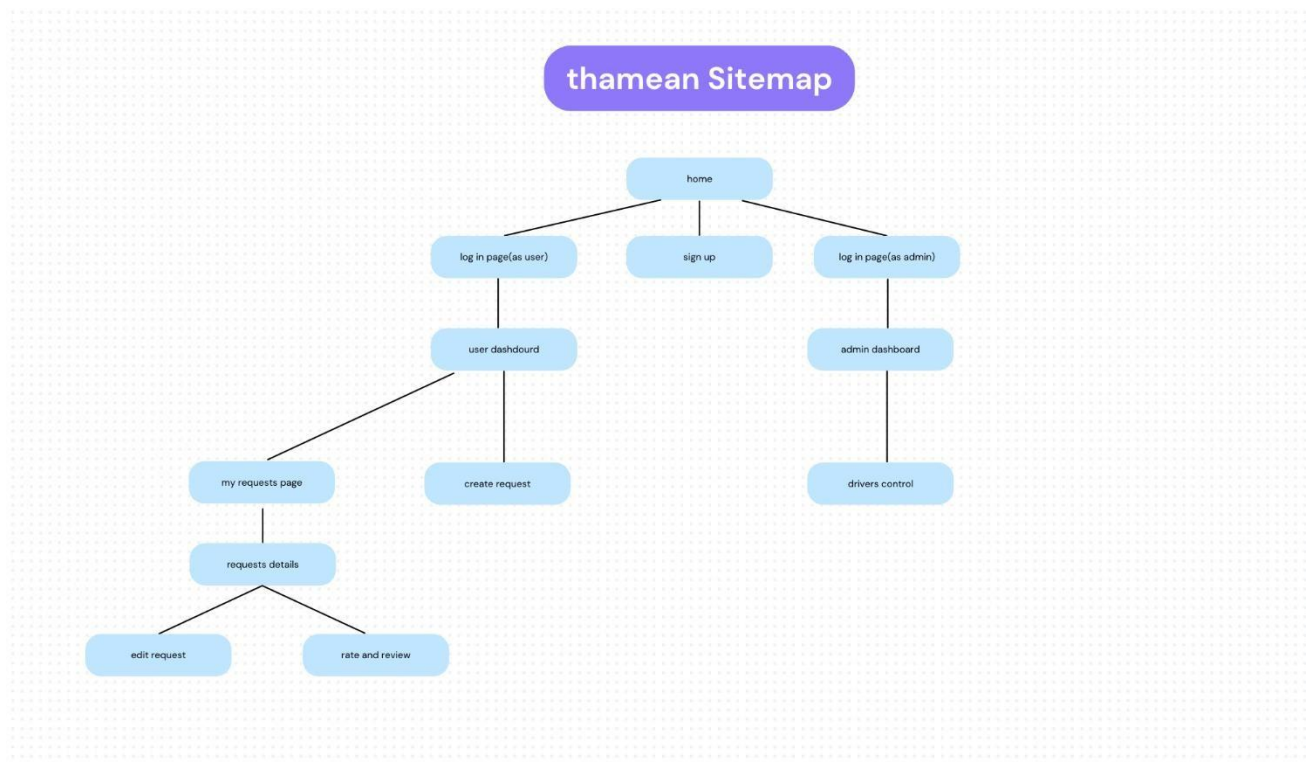


Figure 9: sitemap

4.5.1 Github link:

<https://github.com/shahd351/SoftWarePHP1.git>

4.5.2 Jira link:

<https://2026-it320320-section1910-g4.atlassian.net/jira/software/projects/SW/boards/2/backlog>

5 Chapter 5: Testing

In this chapter, we document the software testing activities performed for the Thameen system.

Three types of testing were conducted: User Story Acceptance Testing, Integration Testing, and User Acceptance Testing. These testing methods were carried out to ensure that the system meets both functional requirements and user expectations. The goal is to guarantee a robust, secure, and userfriendly product that fulfills the needs of individuals and businesses transporting valuable items within Riyadh.

5.1 User Story Acceptance Testing

We will begin with User Story Acceptance Testing, where each feature is evaluated against the agreed-upon user stories to ensure user needs fulfillment from the user perspective, by making sure that the acceptance criteria is met for each user story.

We first prepared the user stories to be tested, then conducted the testing with the development team in a prepared setting and environment. We then observed the system behavior while executing the test cases to ensure that the acceptance criteria were met.

All user stories were tested, which are related to: register, user login, admin login, driver login, user logout, admin logout, driver logout, add delivery request, view requests, view request details, search by request ID, edit request, delete request, admin add driver, admin delete driver, admin assign request to driver, admin dashboard, driver view assigned request, driver mark as delivered, rate service, submit review, insurance coverage, in addition to non-functional requirements related user stories such as speed and availability. Detailed information about the user stories tested and their acceptance criteria will be provided in the following table.

Table 8: user story acceptance testing

Sprint Number	User story	Acceptance criteria	Test cases(s)	Pass ?	Comments
2	As a user I want to register so that I can	• If I enter a valid Full Name, Phone Number, Date of Birth shows I am 18	1. Go to signup.php page. 2. Fill all fields with valid data (Full Name, New Phone Number, Date of	Yes	-

	access Thameen services and create delivery requests.	years or older, and a Password with a minimum of 8 digits and click Register, then the system should create my account successfully and allow me to access the system by redirecting me to Login. - If I enter a Date of Birth that indicates I am under 18 years old, and click Register, then the system should display an error message stating that I must be at least 18 years old to register. - If I enter a Phone Number that already exists and click Register, then the system should reject the registration and display an error message indicating the phone number is already registered.	Birth 18+ years, Password 8+ characters, matching Confirm Password). 3. Click SIGN UP button. 4. Verify account created successfully and redirected to login page. 5. Try to register with the same phone number again. 6. Verify error message "Phone number already registered". 7. Try to register with Date of Birth indicating age 17 years. 8. Verify error message "You must be at least 18 years old". 9. Try to register with Password length 7 characters. 10. Verify error message "Password must be at least 8 characters".		
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		• If I leave Full Name, Phone Number, or Date of Birth empty, or enter a Password with less than 8 digits, and click Register, then the system should display a validation error and prevent account creation until the inputs are valid.			
2	As a user I want to log in using my phone number and password so that I can access Thameen services and manage my delivery requests.	• If I go to the customer login page and enter a registered Phone Number and the correct Password and click "Login", then the system should authenticate me successfully and allow me to access the system. • If I go to the customer login page and enter an unregistered Phone Number or an incorrect Password and click Login, then the system should	1. Go to loginAsUser.php page. 2. Enter registered Phone Number (0512345678) and correct Password (pass1234). 3. Click Log in button. 4. Verify redirected to home.php successfully. 5. Log out and try to login with correct Phone Number but wrong Password. 6. Verify generic error message "Invalid entries". 7. Try to login with unregistered Phone Number. 8. Verify generic error message "Invalid entries". 9. Leave Phone Number or	Yes	-

		deny access and display a generic error message stating that the Phone Number or password is incorrect, without specifying which one for security reasons. • If I leave the Phone Number or Password empty and click Login, then the system should display a validation error and prevent login until the inputs are provided.	Password empty. 10. Verify validation error appears preventing login.		
2	As an admin, I want to log in to the system so that I access administrative functions.	• If I go to the admin login page and enter a correct username and password, the system allows access to the admin dashboard. • If I go to the admin login page and enter incorrect username or password, then the system displays a generic error message stating that the login credentials are invalid, without	1. Go to loginAsAdmin.php page. 2. Enter username (admin1) and password (admin123). 3. Click Log in button. 4. Verify redirected to admin_dashboard.php successfully. 5. Log out and try to login with correct username but wrong password. 6. Verify error message "Invalid username or password". 7. Try to login with invalid	Yes	-

		specifying which one is wrong for security reasons. Only authorized administrators can access the admin interface.	username. 8. Verify error message "Invalid username or password". 9. Leave username or password empty. 10. Verify validation error appears preventing login.		
2	As a driver, I want to log in so that I can view my assigned deliveries.	- Full Name must exist in Driver table. - National ID must match the Full Name. - If credentials are correct, redirect to driver_dashboard.php. - If credentials are incorrect, display error message "Invalid Name or National ID". - If fields are empty, display validation error and prevent login.	1. Go to loginAsDriver.php page. 2. Enter Full Name (Ahmed Alqahtani) and National ID (1234567890). 3. Click Log in button. 4. Verify redirected to driver_dashboard.php successfully. 5. Log out and try to login with correct Full Name but wrong National ID. 6. Verify error message "Invalid Name or National ID". 7. Try to login with unregistered Full Name. 8. Verify error message "Invalid Name or National ID". 9. Leave Full Name or National ID empty. 10. Verify validation error appears preventing login.	Yes	-

2	As a user, I want to log out from my account so that I can protect my personal and shipment information.	- When the user clicks the "Logout" button, the system ends the session. - The user is redirected to the login page. - The user cannot access account pages after logout unless logging in again.	- Login as user to home.php. - Click Logout button at the top left corner. - Verify redirected to login.html.	Yes	-
2	As an admin, I want to log out from the admin dashboard so that I can protect system data and shipment management information.	- When the admin clicks the "Logout" button, the system ends the admin session. - The admin is redirected to the admin login page. - The admin cannot access dashboard or management pages after logout unless logging in again.	- Login as admin to admin_dashboard.php. - Click Logout icon at the top left corner. - Verify redirected to login.html.	Yes	-
2	As a driver, I want to log out so that I can protect my session.	- When the driver clicks the "Logout" button, the system ends the driver session. - The driver is redirected to the login page.	- Login as driver to driver_dashboard.php. - Click Logout icon at the top left corner. - Verify redirected to login.html.	Yes	-

		The driver cannot access driver dashboard or delivery pages after logout unless logging in again.			
2	As a user I want to add a delivery request so that I can transport valuable items securely within Riyadh.	If I am logged in, select an allowed item type (Jewelry, Cash, or High-value Electronics), choose an item value range (Less than 5,000 SAR, 5,000-10,000 SAR, or More than 10,000), enter a pickup location and drop-off location within Riyadh, enter a 4-digit security code that will be used to open the secure locker, and confirm it, and click "Create Request", then the system should create the request successfully and display a confirmation with a unique Order Number, and show the service price and	- Go to CreateRequest.php page. - Fill all required fields with valid data: item type, item value range, pickup location, drop-off location, and 4-digit security code. - Verify that the service price is displayed based on the selected item type. - Verify that the insurance coverage is shown as up to 10,000 SAR. - Click "Add Request". - Verify that the request is created successfully with a unique Request ID.	Yes	-

		insurance coverage (up to 10,000 SAR) before final submission. - If I leave any required field (item type, item value range, pickup location, or drop-off location) empty and click "Add Request", then the system should display a validation error and prevent request creation until all required inputs are provided. - If I enter a pickup or drop-off location outside Riyadh, then the system should reject the request and display an error message explaining the restriction. - If I select an item value range above 10,000 SAR, then the system should display a message informing me that the insurance	- Verify that the new request appears in My Requests page. - Try to submit the form with one required field empty. - Verify that the system displays a validation error and prevents request creation. - Try to enter pickup or drop-off location outside Riyadh. - Verify that the system rejects the request and displays an error message. - Try to enter an invalid security code that is not 4 digits. - Verify that the system displays an error message. - Select each item type and verify the correct service price: Jewelry: 200 SAR, Cash: 150 SAR, High-value		
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		coverage included in the service is limited to 10,000 SAR. - The system should automatically calculate and display the service price based on the selected item type: - Jewelry: 200 SAR - Cash: 150 SAR High-value Electronics: 120 SAR The price must be shown before final submission, and if I confirm, the request is created with the calculated price. - The security code is stored securely and is never displayed to anyone, including the driver. Only the sender and the receiver know the code. Upon delivery, the receiver must provide this code to the driver to unlock the locker and complete the delivery.	Electronics: 120 SAR.		
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2	As a user, I want to view my request details and delivery status so that I see the progress of my shipment.	• If I log in to my account and open my requests page, then I should be able to view the details of my request. • The system should display the current delivery status of the request (Pending, In Transit, or Delivered). • If I open a specific request, then the system should show the item type, item value range, pickup location, drop-off location, and order number. • The request details and status should be shown in a clear and readable format. • If I open my requests page and have no delivery requests, then the system should display a message indicating that no requests are available.	1. Go to loginAsUser.php page and log in with valid credentials. 2. Navigate to "MyRequests.php" page. 3. Verify that a list of user requests is displayed. 4. Select a specific request. 5. Verify that the RequestDetails.php page is opened. 6. Verify that the following information is displayed correctly: item type, item value range, pickup location, drop-off location, and request ID. 7. Verify that the delivery status is displayed (Pending / In Transit / Delivered). 8. Verify that the details and status are	Yes	-

		Only the owner of the request should be able to view its details and status.	shown clearly and in a readable format. 9. Try opening a request that belongs to the logged-in user and verify it is displayed successfully. 10. Try accessing a request using a different user account. 11. Verify that the system prevents access and does not display the request details. 12. Go to “MyRequests.php” page when there are no requests. 13. Verify that a message is displayed indicating that no requests are available.		
2	As a user, I want to search for my request using	If I enter a valid request ID in the search field, then the system should display	1. Go to loginAsUser.php page and log in with valid credentials.	Yes	-

	a request ID so that I can quickly find a specific request.	the corresponding request details. - If the entered request ID does not exist, then the system should display an error message. - The search results should only show requests that belong to the logged in user.	- Navigate to “MyRequests.php” page. - Locate the search field. - Enter a valid request ID that belongs to the logged-in user. - Click Search. - Verify that the system displays the corresponding request details in RequestDetails.php page. - Verify that the displayed request belongs to the logged-in user. - Enter a request ID that does not exist. - Click Search. - Verify that the system displays an error message (“Request not found”). - Enter a request ID that belongs to another user. - Click Search.		
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			13. Verify that the system does not display the request and shows an error or no result. 14. Leave the search field empty and perform search. 15. Verify that the system handles it properly (shows validation message or no results).		
	As a user I want to edit my delivery request before final submission so that I can correct or update the request details.	If my delivery request is not finally submitted and I click Edit, then the system should display the current delivery request details, including the item type, item value range, and the pickup and drop-off locations within Riyadh. • If the modified details (item type, item value range, pickup location, or drop-off location) and I click	1. Go to MyRequests.php page. 2. Select a request that is not delivered. 3. Click Edit button. 4. Modify the request details with valid data. 5. Click Save button. 6. Check that the success message appears and the updated request is displayed.	Yes	-

		“Save”, and the pickup and drop-off are within Riyadh, then the system should update the delivery request with the new information and display a success message. • If I select an item value range above 10,000 SAR, then the system should display a message informing me that			
2	As a user, I want to delete my delivery request before it is marked as “Delivered” so that I can cancel it if I no longer need the service.	• If I am logged in and viewing a delivery request that is not yet marked as “Delivered”, and I click “Delete”, then the system should ask for confirmation (e.g., “Are you sure you want to delete this request? This action cannot be undone.”). • If I confirm the deletion, the system should permanently remove the request from my list and	1. Go to loginAsUser.php page and log in with valid credentials. 2. Navigate to “MyRequests.php” page. 3. Select a request that is NOT marked as “Delivered”. 4. Click “Delete” button. 5. Verify that a confirmation message is displayed (“Are you sure you want to delete this request? This action cannot be undone.”). 6. Click “Confirm” on the confirmation dialog.	Yes	-

		display a success message (e.g., “Request deleted successfully.”). • If I cancel the confirmation, the system should close the confirmation dialog and keep the request unchanged. • If the delivery request is already marked as “Delivered”, the system should not display the “Delete” option for that request, and the user cannot delete it.	7. Verify that the request is permanently removed from the list. 8. Verify that a success message is displayed (“Request deleted successfully”). 9. Verify that the deleted request no longer appears in “MyRequests.php”. 10. Repeat the steps and click “Delete” again. 11. Click “Cancel” on the confirmation dialog. 12. Verify that the confirmation dialog is closed. 13. Verify that the request is NOT deleted and still appears in the list. 14. Select a request that is marked as “Delivered”. 15. Verify that the “Delete” option is NOT displayed.		
2	As an admin, I want to add driver accounts so that I can assign delivery requests to	• If I enter a valid driver full name, phone number, national ID, and vehicle details and click “Add Driver”, then the system should create the	1. Go to the drivers_control.php page. 2. Try to add a driver with valid Full Name, Phone Number (10 digits), and National ID (10 digits). 3. Verify that the driver is added successfully and	Yes	-

	available drivers.	driver account successfully. • If any required field is empty, then the system should display a validation error message and prevent submission. • If I enter a phone number or national ID that already exists, then the system should reject the request and display an error message. • The newly added driver should appear in the drivers list. • Only logged-in administrators can add drivers.	appears in the drivers list with status "Available". 4. Try to add a driver with an empty Full Name field. 5. Verify that the system displays a validation error and prevents submission. 6. Try to add a driver with an empty Phone Number field. 7. Verify that the system displays a validation error and prevents submission. 8. Try to add a driver with an empty National ID field. 9. Verify that the system displays a validation error and prevents submission. 10. Try to add a driver with a phone number that is less than 10 digits. 11. Verify that the system displays an error message and rejects the input. 12. Try to add a driver with a national ID that is less than 10 digits. 13. Verify that the system displays an error message and rejects the input.		
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			14. Try to add a driver with a phone number that already exists. 15. Verify that the system rejects the request and displays an error message. 16. Try to add a driver with a national ID that already exists. 17. Verify that the system rejects the request and displays an error message.		
2	As an admin, I want to delete a driver account so that I can remove inactive or incorrect driver records.	- If I select a driver and click “Delete”, the system should show a confirmation message. - If I confirm, the driver should be removed from the system. - After deletion, the driver should no longer appear in the list.	1. Go to the drivers_control.php page. 2. Try to delete a driver by clicking the “Delete” button. 3. Verify that a confirmation message is displayed. 4. Confirm the deletion. 5. Verify that the driver no longer appears in the drivers list. 6. Try to delete a driver and click “Cancel” in the confirmation message. 7. Verify that the driver is NOT deleted and still appears in the drivers list.	Yes	-
2	As a driver, I want to view my assigned request so	- The driver must be logged in to access the dashboard. - The system should	1. Login as driver (Ahmed Alqahtani) to driver_dashboard.php. 2. Verify the dashboard	Yes	-

	that I can see the delivery details before starting the trip.	display the assigned request with Request ID, Item Type, Pickup Location, Dropoff Location, and Status. - If the driver has no assigned request, the system should display a message indicating no active deliveries. - Only requests with status "In Transit" assigned to this specific driver should be shown.	displays the assigned request details (Request ID, Item Type, Pickup, Dropoff, Status). 3. Verify the Status is displayed as "In Transit" with appropriate color. 4. Login as driver who has no assigned request (Yousef Almutairi). 5. Verify message appears: "No active deliveries assigned to you at this time." 6. Verify that requests assigned to other drivers are not visible to this driver.		
2	As a driver, I want to mark delivery as completed so that I can finish the delivery and become available for new requests.	- The driver must be logged in and have an active request. - The "Mark as Delivered" button should only appear when there is an active request. - When clicked, the system should ask for confirmation before proceeding. - After confirmation, the request status should change from "In Transit" to	1. Login as driver with active request (Ahmed Alqahtani). 2. Verify "Mark as Delivered" button is visible. 3. Click the button and confirm the action when prompted. 4. Verify request status changes from "In Transit" to "Delivered". 5. Check the driver status in the database changes from "Busy" to "Available". 6. Verify the "Mark as Delivered" button	Yes	-

		"Delivered". - The driver status should change from "Busy" to "Available". - The button should disappear after completion. - The driver can now receive new requests.	disappears from the dashboard. 7. Create a new delivery request from a user account. 8. Verify that the same driver can receive a new request since status is now "Available". 9. Try to mark a delivery that is not assigned to the logged-in driver. 10. Verify the action is prevented or the request is not visible.		
	As a user, I want to rate the delivery service after successful delivery so that I can share my satisfaction level.	- If the delivery status is "Delivered", the user can select a rating from 1 to 5 stars. - The system should save the rating successfully. - The rating should be included in the weekly average rating displayed in the admin dashboard. - A confirmation message should be displayed after submission.	1. Go to Rate and Review page for a delivered request. 2. Select a rating from 1 to 5 stars. 3. Click Submit button. 4. Check that the rating is saved successfully. 5. Check that a confirmation message appears.	Yes	-
	As a user, I want to submit a	If the delivery status is "Delivered", the	1. Go to Rate and Review page for a delivered request.	Yes	-

	written review after the delivery is marked as Delivered so that I can provide detailed feedback to the admin.	user can submit a written review. - The review text field must not be empty. - The system should save the review successfully. - The review should be displayed in the admin dashboard under the reviews section. - A confirmation message should be displayed after submission.	2. Enter a written review in the review text field. 3. Click Submit Review button 4. Check that the review is saved successfully. 5. Check that a confirmation message appears.		
2	As an admin, I want to view a dashboard that shows a weekly summary of delivery requests so that I can monitor the system.	- The dashboard page is accessible only to logged-in admins. - The dashboard displays the current week summary automatically. - The dashboard shows: - Total number of delivery requests created this week - Number of requests in each status	1. Go to the loginAsAdmin.php page. 2. Enter valid admin username and password. 3. Click the “Login” button. 4. Verify that the admin is redirected to the admin_dashboard.php page. 5. Verify that the dashboard displays the current week summary. 6. Verify that the dashboard shows the total number of delivery requests created this week.	Yes	-

		(Pending / In Transit / Delivered) • The dashboard displays all ratings and written reviews submitted this week. • The dashboard shows the average rating for this week.	7. Verify that the dashboard shows the number of requests in each status: Pending, In Transit, and Delivered. 8. Verify that the dashboard displays the ratings and written reviews submitted this week. 9. Verify that the dashboard shows the average rating for this week. 10. Log out from the admin dashboard. 11. Try to access the admin_dashboard.php page without logging in. 12. Verify that the dashboard is not accessible without admin login.		
2	As a user, I want my delivery request to include insurance coverage so that my valuable items are protected during	-The system should automatically include insurance coverage up to 10,000 SAR for every delivery request. -The insurance coverage amount should be fixed.	1. Go to "CreateRequest.php" page. 2. Fill all required request details and submit the request. 3. Verify that the system automatically assigns insurance coverage of 10,000 SAR. 4. Verify that the insurance amount is fixed and cannot be modified by the user.	Yes	-

	transportation .	-The insurance information should be saved with the request. -The insurance coverage information should be displayed within the request details.	5. Try to edit the insurance amount. 6. Verify that the system does not allow changing the insurance value. 7. Submit the request successfully. 8. Go to "MyRequests.php" page. 9. Open the created request details "RequestDetails.php". 10. Verify that the insurance information is displayed correctly within the request details. 11. Verify that the insurance value is stored correctly in the system. 12. Verify that insurance is not applied unless request is successfully created.		
2	As a user, I want the system pages to load quickly so that I can track and create delivery requests	- Main pages load within 5 seconds. - Shipment status page loads within 5 seconds after request. - The system handles multiple users without crashing.	1. Go to the login page. 2. Log in with valid user credentials. 3. Open the home page. 4. Verify that the home page loads within 5 seconds. 5. Go to CreateRequest.php page.	Yes	-

	without delay.		- Verify that the page loads within 5 seconds. - Go to MyRequests.php page. - Verify that the requests list loads within 5 seconds. - Open a shipment/request details page. - Verify that the shipment status and request details load within 5 seconds. - Create or open requests from more than one user account. - Verify that the system handles multiple users without crashing or delay.		
2	As a user, I want the system to be available whenever I access it so	- The website is available 24/7. - If the server fails, the system restores service quickly. - Users can access	- Open the system website in the browser. - Verify that the website is accessible.	Yes	-

	that I can check shipment status at any time.	shipment tracking at any time.	- Log in with valid user credentials. - Go to MyRequests.php page. - Open a "RequestDetails.php" page. - Verify that the shipment status can be accessed successfully. - Refresh the page and verify that the system is still available. - Log out and log in again. - Verify that the system allows access again without errors. - If the server is restarted, open the website again. - Verify that the system restores service and pages are accessible again.		
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5.2 Integration Testing

This chapter presents the testing of the final version of the Thameen system. The purpose of this testing is to ensure that all system features function correctly and meet the specified requirements from both system and user perspectives.

Different types of testing were conducted, including User Story Acceptance Testing, Integration Testing, and User Acceptance Testing. We first prepared the test scenarios and test cases based on the system features, then executed them in a controlled environment. During the testing process, we observed the system behavior and verified that all components work correctly and that the expected results were achieved.

5.2.1 Integration Hierarchy

This section presents the integration hierarchy, illustrating the execution flow between the main system components. The diagram demonstrates how different functionalities are connected and interact across three primary user roles: user, driver, and admin.

For the user role, the flow begins with registration or login, followed by navigation to the home page. From there, the user can create a request, view requests, search by ID, or log out. Additionally, the user can access request details and perform actions such as editing, deleting, and submitting ratings and reviews.

For the driver role, the flow starts with logging into the driver dashboard, where the driver can view assigned requests and mark them as delivered.

For the admin role, the flow shows logging into the admin dashboard, managing drivers (adding, deleting, and assigning drivers), and viewing system data through the dashboard.

This integration hierarchy provides a clear understanding of how system components are connected and how execution flows seamlessly between different modules.

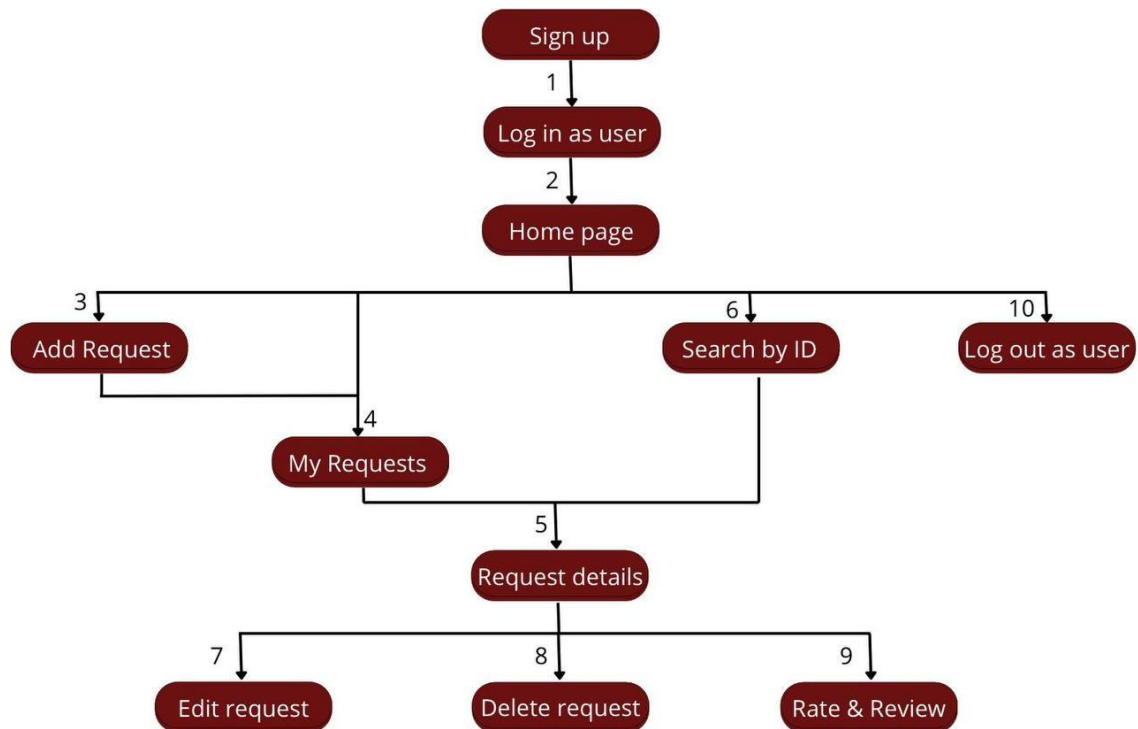


Figure 10: integration hierarchy as a user

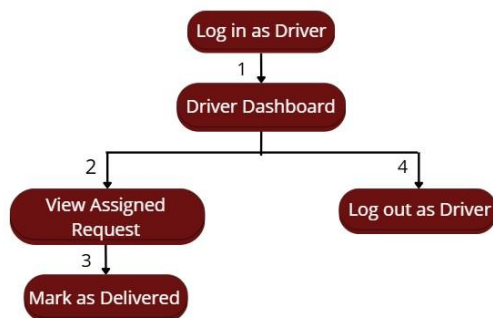


Figure 11: integration hierarchy as a driver

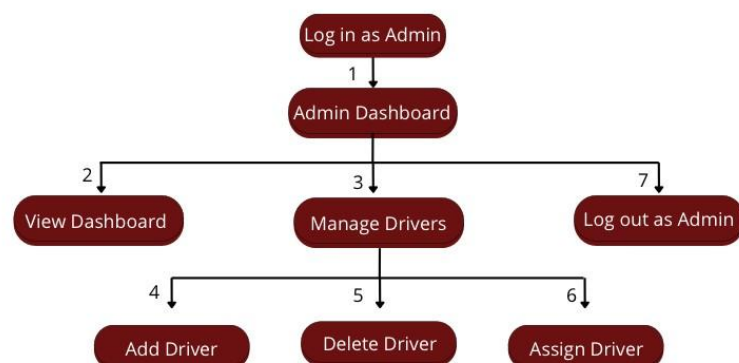


Figure 12: integration hierarchy as an admin

5.2.2 Integration Testing Plan

This section presents the integration testing plan, which is designed to verify the interaction between different system components after integration. The purpose of this testing is to ensure that the system modules work together correctly and that data flows smoothly between them.

The integration scenarios focus on key system functionalities, including user authentication, viewing requests, and managing requests (such as editing, rating, and reviewing). These scenarios represent typical user interactions within the system and demonstrate how multiple components are combined to perform complete tasks.

Each test case outlines a sequence of actions performed by the user and checks whether the system responds as expected. The results confirm whether the integration between components is successful and whether the system behaves correctly in real usage scenarios.

Table 9: integration testing scenarios

System components	New component	Test case	Pass?	Comments
-Logging as user in the system	Viewing My Requests	1. A user logs in to the system successfully. 2. The logged-in user navigates to the "MyRequests.php" page.	Yes	
		3. The system displays all requests belonging to the user.		

-Logging as user in the system -Viewing My Requests	Managing request (Edit / Rate & Review)	1. A user logs in to the system successfully. 2. The logged-in user navigates to the “My Requests” page. 3. The system displays all requests belonging to the user. 4. The logged-in user selects a request from the list of available requests. 5. If the request is not delivered, the user clicks the “Edit” option and updates the request details successfully. 6. If the request status is “Delivered”, the user selects a	Yes	
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		rating and writes a review. 7. The user submits the rating and review. 8. The system saves the changes successfully and displays a confirmation message.		
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5.3 User Acceptance Testing

The last testing to be performed is the user acceptance testing, which allows real users from the targeted audience to test the application in real-world scenarios, ensuring that it meets their requirements and performs effectively in real life situations, and it is also used to check if the system fulfills business requirements.

To carry out this test, we recruited five participants who met the following criteria: adults aged 18 years or older, residents of Riyadh, and had basic familiarity with web-based services. The participants included both individuals and shop owners who regularly transport valuable items such as jewelry, cash, or electronics.

The test was conducted in a quiet room with a laptop provided to each participant. Prior to the test, we gave the users three minutes to familiarize themselves with the Thameen web application. Following this, the test was performed based on a scenario that covered all the main functionalities of the system, including:

- Registering a new account
- Logging in to the system
- Creating a new delivery request
- Viewing the list of requests (My Requests)

- Viewing request details
- Searching for a request by Request ID
- Editing a request before final submission
- Deleting a request before it is marked as Delivered
- Submitting a rating and review after delivery
- Logging out from the system

After completing the tasks, participants were asked to fill out a questionnaire consisting of ten questions, where they evaluated their experience and pointed out any functional issues or potential enhancements. The questionnaire focused on ease of use, interface clarity, functionality of each feature, and overall satisfaction with the system.

Lastly, we analyzed the feedback to identify usability problems and areas for improvement. The results of the questionnaire, along with the demographic information of the participants, are presented in the following sections. The full questionnaire responses and raw data are provided in Appendix D.

5.3.1 Demographics of participants

In this sub-section, we will represent the participants' demographics. This includes the information we have gathered from the users regarding their age, gender, and user type. We recruited five participants to take part in the User Acceptance Testing. The participants were selected based on the target user criteria: adults aged 18 years or older, residents of Riyadh, and familiar with using web-based services. The group included both individual users and shop owners who regularly need to transport valuable items such as jewelry, cash, or electronics.

We will summarize the demographics results as follows:

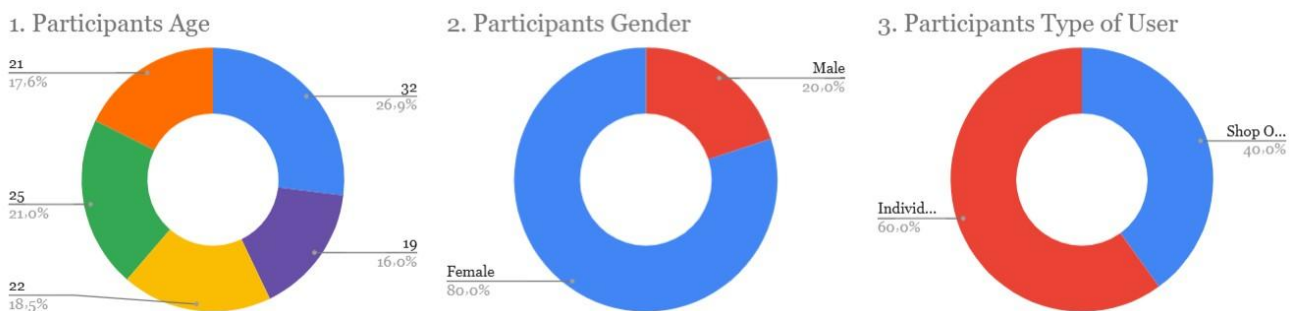


Figure 13: participants demographics(age, gender, and user type)

Age Distribution:

The participant demographics show that the largest age group is 32 years old, representing 26.9% of participants, followed by 25 years at 21.0%, 21 years at 17.6%, 22 years at 18.5%, and 19 years at 16.0%. (See Figure 13)

Gender Distribution:

The participant demographics show a gender distribution with approximately 80% female and 20% male. This indicates a higher representation of female participants in the testing group. (See Figure 13)

User Type Distribution:

The participant demographics show that 60.0% of participants were individual users, while 40.0% were shop owners. This indicates that the majority of participants were individuals who occasionally transport personal valuable items. (See Figure 13)

These graphical representations provide a clear view of the demographics of the participants involved in the testing.

5.3.2 Questionnaire

In this sub-section, we present the results of the user questionnaire. The results reflect participants' feedback on the system's features and overall usability. The data is analyzed to understand how users interacted with the system and to identify areas of strength and potential improvement.

Results:

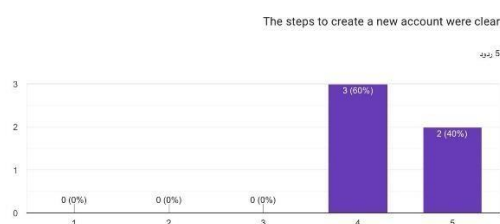


Figure 14: question 1

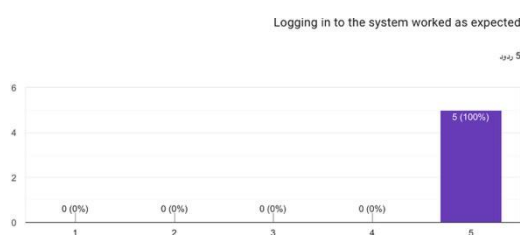


Figure 15: question 2

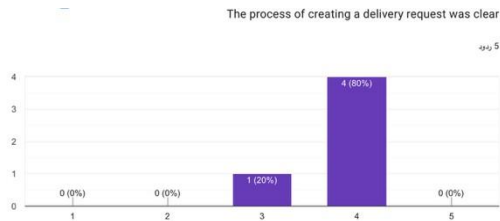


Figure 16: question 3

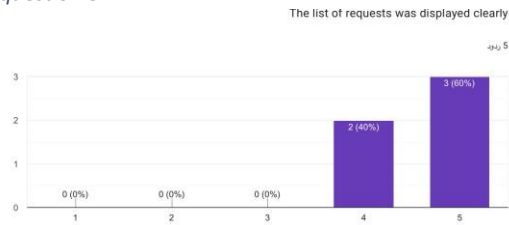


Figure 17: question 4

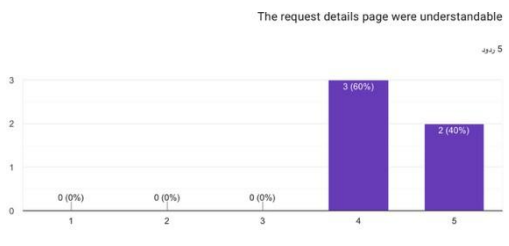


Figure 18: question 5

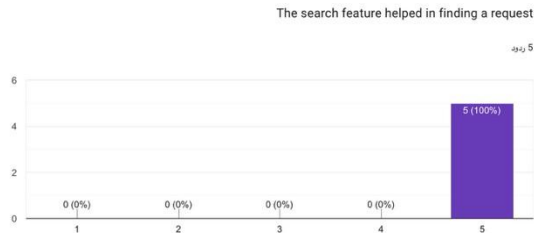


Figure 19: question 6

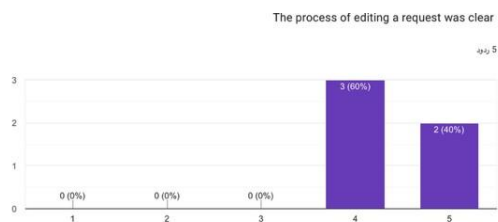


Figure 20: question 7

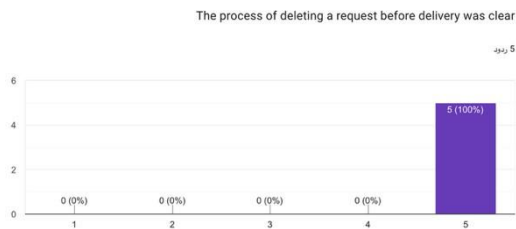


Figure 21: question 8

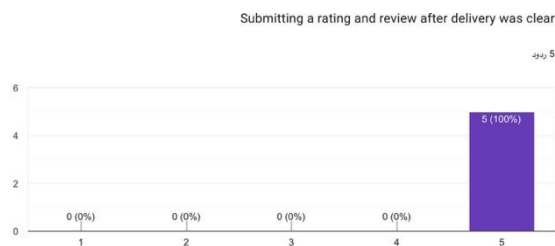


Figure 22: question 9

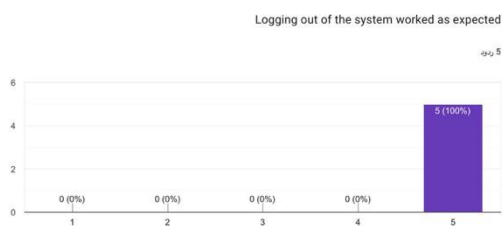


Figure 23: question 10

Q#	Question	P1	P2	P3	P4	P5	Mean
1	Account creation steps were clear	4	4	5	4	5	4.4
2	Login worked as expected	5	5	5	5	5	5.0
3	Creating a request was clear	4	4	3	4	4	3.8
4	Requests list was clear	5	4	5	5	4	4.6
5	Request details were clear	4	5	4	4	5	4.4
6	Search worked well	5	5	5	5	5	5.0
7	Editing a request was clear	4	4	5	4	5	4.4
8	Deleting a request was clear	5	5	5	5	5	5.0

9	Rating submission was clear	5	5	5	5	5	5.0
10	Logout worked as expected	5	5	5	5	5	5.0

The steps to create a new account were rated positively by all participants. Most of them (60%) selected a rating of 4, while the remaining 40% selected 5. This indicates that the registration process was generally clear.

Logging into the system worked as expected for all participants. A full 100% selected a rating of 5, showing that the login functionality was reliable and did not cause any issues.

The process of creating a delivery request received slightly lower ratings compared to other features. While 80% of participants rated it as 4, 20% rated it as 3. This suggests that some users may have experienced minor confusion during this process.

The display of the requests list was clear and well understood by users. A total of 60% of participants selected 5, while 40% selected 4, indicating that the interface was easy to follow.

The request details page was also rated positively. Around 60% of participants gave a rating of 4, while 40% gave a rating of 5, showing that the information presented was understandable.

The search feature performed very well, with 100% of participants selecting a rating of 5. This confirms that users were able to find requests easily using the search functionality.

The process of editing a request was rated positively as well. About 60% of participants selected 4, while 40% selected 5, indicating that the feature was generally clear.

The process of deleting a request before delivery was rated very highly. All participants (100%) selected 5, showing that the feature was clear and easy to use.

Submitting a rating and review after delivery also received perfect ratings. All participants (100%) selected 5, indicating that this feature was straightforward and easy to complete.

Finally, logging out of the system worked as expected for all users. A total of 100% of participants selected 5, confirming that the logout process was simple and reliable.

Overall, the results reflect a high level of user satisfaction, as participants were able to interact with the system features smoothly and effectively.

5.4 Discussion

This section provides an interpretation of the testing results presented in the previous sections, including User Acceptance Testing (UAT) and Non-Functional Requirements (NFR). It also reflects on the overall system development process, design decisions, and implementation outcomes.

Results (UAT and NFR):

The results of the User Acceptance Testing indicate a high level of user satisfaction with the Thameen system. Most features received ratings between 4 and 5, with several

functionalities such as login, search, deletion, rating submission, and logout achieving a perfect score of 5.0. This demonstrates that these features are reliable, intuitive, and function as expected in real-world scenarios.

However, slightly lower ratings were observed in the process of creating a delivery request (mean = 3.8). This suggests that some users experienced minor confusion when entering request details or understanding the required inputs. This indicates that the request creation process, while functional, may require improvements in usability and clarity.

In addition to UAT results, the Non-Functional Requirements (NFR) were also satisfied. The system demonstrated good performance, with main pages loading within the expected time limits. Furthermore, the system showed high availability, allowing users to access services and track shipments at any time without interruption.

Reflection on Design & Development:

From a design perspective, the system successfully provides a clear and structured user interface, allowing users to navigate between pages such as “My Requests”, “Request Details”, and “Create Request” with ease. The separation of functionalities based on user roles (user, driver, and admin) also enhances system organization and security.

During development, implementing validation rules (such as age restriction, password length, and required fields) and access control mechanisms (ensuring users can only view their own data) contributed significantly to system reliability and security.

Additionally, the integration between system components (authentication, request management, and driver assignment) worked effectively, as confirmed by the integration testing results.

Reflection on Process:

The development process followed a structured approach, starting from requirement analysis, followed by design, implementation, and testing phases. Dividing the system functionalities among team members (user, driver, and admin modules) helped in managing the workload efficiently and ensured parallel development.

Continuous testing throughout the development process (unit, integration, and acceptance testing) helped identify issues early and improve the overall system quality.

Final evaluation:

Overall, the system evaluation can be considered successful. The high ratings across most features indicate that the system meets its functional requirements and provides a satisfactory user experience. The absence of critical errors during testing also confirms that the system is stable and ready for deployment.

Despite the positive results, there are several areas for improvement:

- Simplifying the delivery request creation process by improving form design and adding clearer instructions or hints.
- Enhancing user interface elements to further improve clarity and reduce user confusion.
- Adding real-time validation feedback to guide users.
- Improving error messages to be more descriptive and helpful.
- Expanding testing to include a larger and more diverse group of participants for more reliable results.
- Enhancing system performance and responsiveness, especially under higher user loads.

Future Enhancements:

The system can be enhanced by adding advanced features such as real-time tracking of deliveries, notifications for status updates, and integration with payment systems. Additionally, improving scalability and security mechanisms will ensure the system can handle real-world deployment scenarios.

6 Chapter 6: Conclusion and Future Work

This chapter summarizes the outcomes of the Thameen project and reflects on the overall development process.

The Thameen project was developed to address the need for a secure and reliable solution for transporting high-value items such as jewelry, cash, and expensive electronics. The project followed

a structured development process that began with identifying the problem and analyzing existing solutions, followed by gathering user requirements to ensure that the system meets real user needs.

These requirements were then translated into clear system designs, including use case diagrams, database structure, and user interfaces.

The system was implemented as a web-based application that allows users to manage delivery requests efficiently, while also ensuring security through features such as confidential access codes, insurance coverage, and controlled request handling. Different testing techniques were applied to verify that the system functions correctly, integrates properly, and provides a usable experience for end users.

The final system demonstrates an effective solution that improves organization, security, and trust in transporting valuable items. The following sections present the impact of the system, the challenges faced during development, the system limitations, the main contributions, and possible future improvements.

Global and Local Impact:

Thameen has a positive local impact because it supports individuals and shops in Riyadh who need a secure and organized way to transport valuable items such as jewelry, cash, and expensive electronic devices. The system helps increase trust between customers and the transportation service by providing structured request management, insurance coverage, status updates, and secure locker access using a confidential code.

On a global level, Thameen supports the growing need for technology-based secure transportation solutions. As digital services continue to expand, systems that improve safety, accountability, and transparency in transporting valuable items can contribute to higher service quality standards and encourage more reliable delivery practices.

Problems and Challenges Encountered:

One of the main challenges faced during the development of Thameen was transforming the collected requirements into clear and consistent system designs, including use case diagrams, database structure, and interface design. Ensuring consistency across all components required careful planning and continuous refinement. In addition, integrating different system roles such as

administrators, drivers, and users within the same workflow was technically complex, as it required smooth coordination and communication between all components, which led to simplifying some interactions in the final implementation.

Limitations of the System:

Although Thameen provides a secure transportation solution, the system has some limitations. The service is limited to Riyadh city only and does not support other cities or regions. The system also supports only the English language. In addition, Thameen focuses only on transporting valuable items such as jewelry, cash, and high-value electronics, and it does not support transporting living creatures, weapons, or antique artifacts.

Some advanced features are outside the current scope, such as real-time GPS tracking, online payment integration, and expansion to other cities. Also, the system responsibility ends once the customer confirms receiving the shipment.

The Main Contribution of the Project:

The main contribution of Thameen is providing a specialized web-based transportation management system for valuable items. Unlike general delivery services, Thameen focuses on secure delivery by using dedicated vehicles with locked safes, confidential access codes known only to the sender and receiver, insurance coverage up to 10,000 SAR, and clear delivery statuses.

The system also allows users to create, view, edit, delete, rate, and review delivery requests. For administrators, the system provides driver management and a dashboard that displays weekly delivery summaries, ratings, and reviews. This makes Thameen a more organized and trustworthy solution for valuable item transportation.

Future Work:

In the future, Thameen can be improved by adding real-time GPS tracking so users can monitor the exact location of their shipments. Online payment integration can also be added to allow customers to pay directly through the system. Another future improvement is expanding the service to other cities in Saudi Arabia.

The system can also support Arabic language to make it easier for more local users. In addition, mobile application support can be developed to provide faster and more convenient access. More advanced admin reports can also be added to help administrators analyze delivery performance, customer satisfaction, and driver activity more effectively.

7 References

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8 Appendix

8.1 Appendix A: Interviews

This appendix contains the interview questions used during the study and the full transcription of all participants responses.

Interviews Questions:

1. Have you ever needed to transport valuable items such as jewelry, cash, or expensive electronic devices within the city?
2. What concerns do you have when transporting valuable items?
3. How would you prefer to request a transportation service for valuable items?

4. What features would make you feel more comfortable when transporting valuable items?
5. Would you prefer using a specialized transportation service for valuable items instead of general delivery services? Why?

Interviews Transcriptin:

Interview 1:

Figure 24: interview 1

Interview Outline (1)	
Interviewee: Norah	Interviewer: Albatool
Location: Online	Appointment Date: 2026/2/26 Start Time: 12:00 End Time: 12:31
Objectives: - Understand the interviewee's experience with transporting valuable items within the city - Identify main concerns related to safety and trust during transportation - Explore user expectations and preferred features in a secure transportation system	Reminders: The interviewee owns a shop and regularly deals with transporting valuable items to customers, making transportation reliability and security critical to her business operations.
Agenda: - Introduction - Brief explanation of the project - Overview of interview - Permission to record - Question 1 - Question 2 - Question 3 - Question 4 - Question 5 - Summary of key points - Questions from the interviewee - Closing	Approximate Time: 2 min 2 min 1 min 0:30 sec 4 min 3 min 4 min 5 min 5 min 2 min 2 min 0:30 sec
General Observation: During the interview, the interviewee was professional and focused, sharing practical insights based on her experience as a shop owner and emphasizing business related concerns and expectations.	

Unresolved Issue, Topics Not Covered: due to time limitations, some details were not fully discussed .	
Interviewee: Norah	Date: 2026/2/26
Questions 1: Have you ever needed to transport valuable items such as jewelry, cash, or expensive electronic devices within the city?	Answer: Yes, I often need to transport valuable items related to my shop, especially items that require careful handling. These items are usually fragile or expensive, which makes transportation a sensitive process, Because of this, I always think carefully about how and with whom these items are transported. Observation: She has regular experience transporting valuable items and is aware of the risks involved.
Questions 2: What concerns do you have when transporting valuable items?	Answer: One concern I have is the risk of mistakes during the delivery process, such as sending the item to the wrong location or mixing it with other shipments. Observation: She is concerned about errors during the delivery process
Questions 3: How would you prefer to request a transportation service for valuable items?	Answer: I prefer requesting the service through an online system that clearly explains the steps of the process, The system should allow me to provide detailed information about the item and give me confirmation that the request has been successfully submitted. Observation: She prefers a clear and structured online system for requesting transportation services.
Questions 4: What features would make you feel more comfortable when transporting valuable items?	Answer: I would feel more comfortable if the service clearly defined how valuable items are handled and stored during transportation, Knowing that the item is placed in a secure and controlled environment and that access to it is limited would increase my confidence in the service. Observation: She values secure storage and controlled access during delivery.
Questions 5: Would you prefer using a specialized transportation service for valuable items instead of general delivery services? Why?	Answer: Yes, I would prefer a specialized service because it focuses specifically on the protection of valuable items, General delivery services do not always consider the special requirements of valuable goods, while a specialized service gives me more confidence and peace of mind. Observation: She prefers specialized services that focus on protection and careful handling.

Interview 2:

Interview Outline (2)	
Interviewee: Faisal	Interviewer: Shahd
Location: Online	Appointment Date: 2026/2/26 Start Time: 11:00 End Time: 11:26
Objectives: • Understand the interviewee's experience with transporting valuable items within the city • Identify main concerns related to safety and trust during transportation • Explore user expectations and preferred features in a secure transportation system	Reminders: The interviewee owns a shop and regularly deals with transporting valuable items to customers, using general delivery services.
Agenda: • Introduction • Brief explanation of the project • Overview of interview • Permission to record • Question 1 • Question 2 • Question 3 • Question 4 • Question 5 • Summary of key points • Questions from the interviewee • Closing	Approximate Time: 1.5 min 2 min 1 min 0:30 sec 3 min 4 min 3 min 3 min 4 min 2 min 1 min 1 min
General Observation: During the interview, the interviewee was confident and knowledgeable, providing detailed responses based on his business experience.	

Figure 25:interview 2

Unresolved Issue, Topics Not Covered: Unfortunately, some details were not fully discussed due to time limitations.	
Interviewee: Faisal	Date: 2026/2/26
Questions 1: Have you ever needed to transport valuable items such as jewelry, cash, or expensive electronic devices within the city?	Answer: Yes, as a shop owner I frequently need to transport valuable items such as jewelry and high value products to customers, This happens on a regular basis, which makes transportation an important part of my daily business operations. Observation: He regularly transports valuable items as part of his business activities.
Questions 2: What concerns do you have when transporting valuable items?	Answer: My main concerns are loss and damage during transportation, I also worry about delays and whether the shipment is properly monitored, especially that the items are expensive and valuable . Observation: He is concerned about security, damage, and shipment monitoring.
Questions 3: How would you prefer to request a transportation service for valuable items?	Answer: I prefer requesting transportation services through an online system or application, This method is more efficient and allows me to submit requests quickly without relying on phone calls or manual coordination. Observation: He prefers a fast and organized digital method for requesting services.
Questions 4: What features would make you feel more comfortable when transporting valuable items?	Answer: secure handling, clear delivery status updates, insurance coverage, would make me feel more comfortable. These features help reduce risks and increase trust in the service. Observation: He values security features that reduce risk and increase reliability.
Questions 5: Would you prefer using a specialized transportation service for valuable items instead of general delivery services? Why?	Answer: Yes, I would prefer using a specialized transportation service because it focuses more on security and reliability. This is important for protecting valuable items and maintaining customer trust in my business. Observation: He prefers specialized services over general delivery options for business use.

Interview 3:

Interview Outline (3)	
Interviewee: Reem	Interviewer: Amal
Location: Online	Appointment Date: 2026/2/25 Start Time: 10:00 End Time: 10:28
Objectives: • Understand the interviewee's experience with transporting valuable items within the city • Identify main concerns related to safety and trust during transportation • Explore user expectations and preferred features in a secure transportation system	Reminders: The interviewee has previous experience transporting valuable personal items and is familiar with general delivery services.
Agenda: • Introduction • Brief explanation of the project • Overview of interview • Permission to record • Question 1 • Question 2 • Question 3 • Question 4 • Question 5 • Summary of key points • Questions from the interviewee • Closing	Approximate Time: 2 min 2 min 1:30 min 0:30 sec 2 min 3 min 3 min 4 min 5min 2 min 2 min 1 min
General Observation: During the interview, the interviewee was attentive and actively engaged in the discussion, She answered the questions clearly and shared relevant personal experiences.	

Figure 26: interview 3

Unresolved Issue, Topics Not Covered: some details were not fully discussed due to time limitations.	
Interviewee: Reem	Date: 2026/2/25
Questions 1: Have you ever needed to transport valuable items such as jewelry, cash, or expensive electronic devices within the city?	Answer: Yes, I have needed to transport valuable items several times, especially jewelry and electronic devices. In most cases, I relied on personal arrangements or general delivery services, but I always felt concerned about the safety of the items during transportation. Observation: She has previous experience transporting valuable items.
Questions 2: What concerns do you have when transporting valuable items?	Answer: My main concern is damage, I also worry about not knowing who is handling the item and whether the delivery process is properly monitored. Observation: She is mainly concerned about security and proper handling.
Questions 3: How would you prefer to request a transportation service for valuable items?	Answer: I would prefer to request the service through an online system or web application, It should be simple to use and allow me to submit all required details clearly. Observation: She prefers using an online system to request the service.
Questions 4: What features would make you feel more comfortable when transporting valuable items?	Answer: Features like secure handling, limited access to shipment details, clear delivery status updates, and confirmation upon delivery would make me feel more comfortable, Knowing that the item is insured would also increase my confidence in the service. Observation: She feels more confident with clear tracking and secure handling.
Questions 5: Would you prefer using a specialized transportation service for valuable items instead of general delivery services? Why?	Answer: Yes, I would prefer a specialized service because it focuses on security and reliability, General delivery services are not designed specifically for valuable items, while a specialized service would provide better protection and peace of mind. Observation: She prefers a specialized service over general delivery services.

Figure 27: interview 4

Interview 4:

Interview Outline (4)	
Interviewee: Rawan	Interviewer: Lina
Location: Online	Appointment Date: 2026/2/27 Start Time: 10:00 End Time: 10:29
Objectives: - Understand the interviewee's experience with transporting valuable items within the city - Identify main concerns related to safety and trust during transportation - Explore user expectations and preferred features in a secure transportation system	Reminders: The interviewee is an individual user who occasionally needs to transport personal valuable items and has limited experience with specialized transportation services.
Agenda: - Introduction - Brief explanation of the project - Overview of interview - Permission to record - Question 1 - Question 2 - Question 3 - Question 4 - Question 5 - Summary of key points - Questions from the interviewee - Closing	Approximate Time: 2 min 2 min 1:30 min 0:30 sec 4 min 3 min 3 min 4 min 4 min 2 min 2 min 1 min
General Observation: During the interview, the interviewee was thoughtful and reflective, sharing personal concerns and expectations.	

Unresolved Issue, Topics Not Covered: Some aspects were not discussed in detail due to time limitations.	
Interviewee: Rawan	Date: 2026/2/27
Questions 1: Have you ever needed to transport valuable items such as jewelry, cash, or expensive electronic devices within the city?	Answer: Yes, but not frequently, I have needed to transport items like personal jewelry or electronics and sometimes money on special occasions, such as moving or sending items to family members, Because these items are valuable , I usually feel anxious during the process. Observation: She occasionally transports personal valuable items and associates the process with emotional concern.
Questions 2: What concerns do you have when transporting valuable items?	Answer: My main concern is privacy, I worry about who has access to my items and whether personal information about the shipment is shared unnecessarily. Observation: She is concerned about privacy and limited access to personal shipment details.
Questions 3: How would you prefer to request a transportation service for valuable items?	Answer: I would prefer a simple mobile application that does not require many steps, The process should be clear and not overwhelming, especially for users who do not use delivery services often. Observation: She prefers a simple and user friendly digital request pocess.
Questions 4: What features would make you feel more comfortable when transporting valuable items?	Answer: I would feel more comfortable if the service feels well organized and professional from start to finish, When the process is clear and consistent, it helps build trust that the item is being handled properly throughout the delivery. Observation: She feels more comfortable when the service appears organized and professional.
Questions 5: Would you prefer using a specialized transportation service for valuable items instead of general delivery services? Why?	Answer: Yes, I would prefer a specialized service if it feels trustworthy and easy to use, Knowing that the service is designed specifically for valuable items would make me feel more confident. Observation: She prefers specialized services when they emphasize trust and simplicity.

8.2 Appendix B: Questionnaire

This appendix includes the questionnaire questions distributed to participants in order to collect feedback about the proposed Thameen system. It also presents a visual representation of the responses obtained.

Questionnaire Questions:

Q1. What type of valuable items would you most likely transport using this service?

- Jewelry
- Cash
- Electronics

Q2. Would insurance coverage increase your trust in the delivery service?

- Yes
- No
- Maybe

Q3. How important is delivery tracking (Pending / In Transit / Delivered)?

- 1- Very Important
- 2
- 3
- 4
- 5- Not Important

Q4. How reasonable do you find allowing users to delete a delivery request anytime before it becomes “Delivered”?

- Very Reasonable
- Reasonable

- Neutral
- Unreasonable
- Very Unreasonable

Q5. How important is the ability to rate the service after delivery?

- Very Important
- Important
- Neutral
- Not Important

Q6. Would you prefer fixed pricing for all deliveries or pricing based on item type and value?

- Fixed price for all deliveries
- Price varies depending on item type/value
- Not sure

Questionnaire Answers:

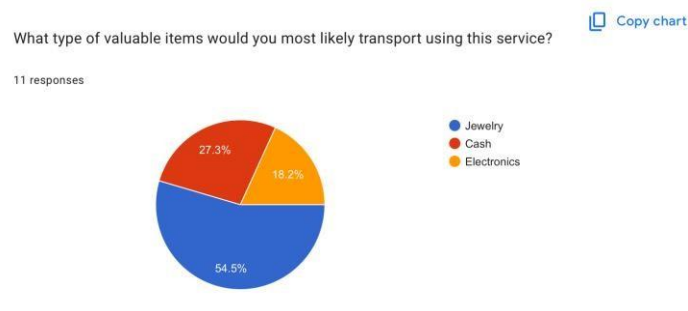


Figure 28: question 1



Figure 29: question 2

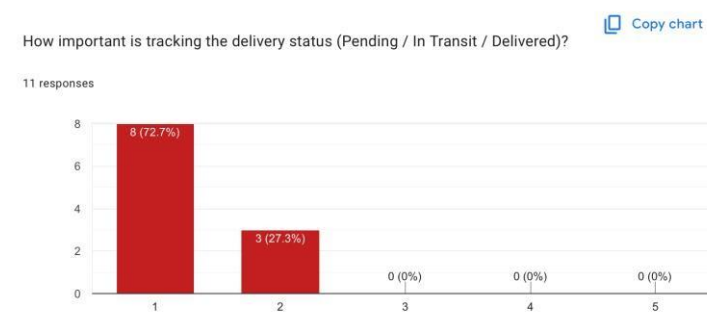


Figure 30: question 3

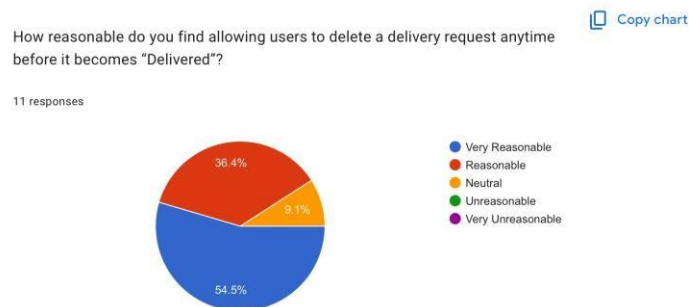


Figure 31: question 4

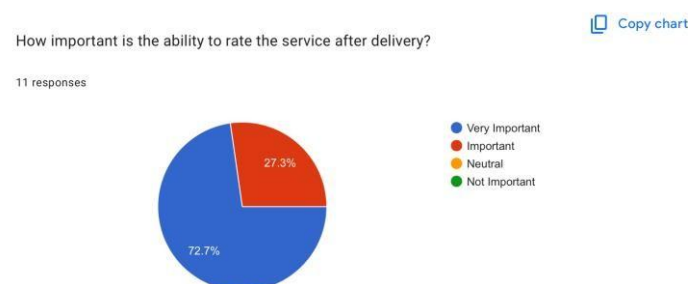


Figure 32: question 5

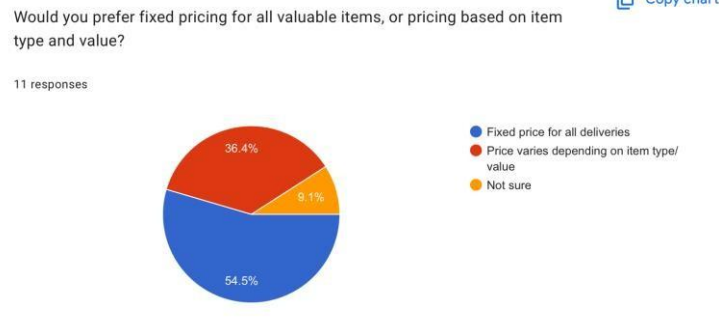


Figure 33: question 6

Questionnaire Responses:

Figure 34: questionnaires responses

	A	B	C	D	E	F	G
1	Timestamp	What type of valuable items would you most likely	Would insurance coverage increase your trust in it?	How important is tracking the delivery status (Penc	How reasonable do you find allowing users to delet	How important is the ability to rate the service afte	Would you prefer fixed pricing for all valuable items
2	02/03/2026 20:57:28	Jewelry	Yes	1 Reasonable	Important	Fixed price for all deliveries	
3	02/03/2026 20:58:26	Electronics	Yes	1 Very Reasonable	Very Important	Not sure	
4	02/03/2026 20:59:08	Jewelry	Yes	1 Neutral	Important	Fixed price for all deliveries	
5	02/03/2026 21:00:36	Cash	Maybe	1 Very Reasonable	Important	Fixed price for all deliveries	
6	02/03/2026 21:01:24	Jewelry	Yes	1 Very Reasonable	Very Important	Fixed price for all deliveries	
7	02/03/2026 21:02:37	Cash	Yes	1 Reasonable	Very Important	Price varies depending on item type/value	
8	02/03/2026 21:04:10	Electronics	Yes	1 Reasonable	Very Important	Price varies depending on item type/value	
9	02/03/2026 21:05:19	Jewelry	Yes	2 Reasonable	Very Important	Fixed price for all deliveries	
10	02/03/2026 21:08:12	Cash	Yes	2 Very Reasonable	Very Important	Fixed price for all deliveries	
11	02/03/2026 21:10:16	Jewelry	Yes	2 Very Reasonable	Very Important	Price varies depending on item type/value	
12	02/03/2026 21:13:36	Jewelry	Yes	1 Very Reasonable	Very Important	Price varies depending on item type/value	
13							

Responses Link:

<https://docs.google.com/spreadsheets/d/1vIBZ7o tkVIF4K4tnvDz qg XWsN tl4ggURVzyzVS8/edit?usp=drivesdk>

8.3 Appendix C: Jira

This appendix presents the use of Jira, a widely used project management tool that was utilized in our project to organize and manage the development process. Jira helps teams track tasks, monitor progress, and collaborate effectively during the project lifecycle. In our project, Jira was mainly used to manage the Product Backlog and organize the work into Sprints following the Scrum methodology.

The Product Backlog contains a list of prioritized tasks and user stories that represent the features and requirements of the system. These tasks are continuously updated and refined throughout the development process to ensure that the team focuses on delivering the most valuable functionalities.

The Sprint represents a short, fixed development period in which selected backlog tasks are implemented and completed by the team. During each sprint, team members work on assigned tasks and track their progress using Jira.

Spaces

2026-IT320--section1910-G4

Summary Timeline Backlog Board Calendar List Forms Development Code More 3 +

Search backlog Filter

Backlog (18 work items) 47 0 0 Create sprint

As a user I want to register so that I can access Thameen services and create delivery req...	TO DO	3	
As a user I want to log in using my phone number and password so that I can access Tha...	TO DO	2	
As an admin, I want to log in to the system so that I access administrative functions.	TO DO	2	
as a user, I want to log out from my account so that I can protect my personal and shipme...	TO DO	2	
As an admin, I want to log out from the admin dashboard so that I can protect system data...	TO DO	2	
As a user I want to add a delivery request so that I can transport valuable items securely ...	TO DO	5	
As a user, I want to view my request details and delivery status so that I see the progress ...	TO DO	2	
As a user, I want to search for my request using a request ID so that I can quickly find a s...	TO DO	3	
As a user I want to edit my delivery request before final submission so that I can correct o...	TO DO	3	
As a user, I want to delete my delivery request before it is marked as "Delivered" so that I ...	TO DO	2	
As an admin, I want to add driver accounts so that I can assign delivery requests to availa...	TO DO	3	

Figure 35: jira. backlog

Spaces

2026-IT320--section1910-G4

Summary Timeline Backlog Board Calendar List Forms Development Code More 3 +

Search backlog Filter

As a user I want to edit my delivery request before final submission so that I can correct o...	TO DO	3	
As a user, I want to delete my delivery request before it is marked as "Delivered" so that I ...	TO DO	2	
As an admin, I want to add driver accounts so that I can assign delivery requests to availa...	TO DO	3	
As an admin, I want to delete a driver account so that I can remove inactive or incorrect dr...	TO DO	2	
As a user, I want to rate the delivery service after successful delivery so that I can share ...	TO DO	3	
As a user, I want to submit a written review after the delivery is marked as Delivered so tha...	TO DO	3	
As an admin, I want to view a dashboard that shows a weekly summary of delivery reques...	TO DO	3	
As a user, I want my delivery request to include insurance coverage so that my valuable it...	TO DO	3	
As a user, I want the system pages to load quickly so that I can track and create delivery r...	TO DO	2	
As a user, I want the system to be available whenever I access it so that I can check ship...	TO DO	2	

Figure 36: jira backlog sprint 2

Spaces

2026-IT320--section1910-G4

Summary Timeline Backlog Board Calendar List Forms Development Code More 3 +

Search work Filter Group Saved filters

Work	Assignee	Reporter	Priority	Status
SW-3 As a user I want to register so that I can acces...	Shahd aldwhan	lina	Medium	DONE ✓
SW-4 As a user I want to log in using my phone num...	Shahd aldwhan	lina	Medium	DONE ✓
SW-5 As an admin, I want to log in to the system so t...	Albatool	lina	Medium	DONE ✓
SW-6 as a user, I want to log out from my account so ...	Shahd aldwhan	lina	Medium	DONE ✓
SW-7 As an admin, I want to log out from the admin d...	Albatool	lina	Medium	DONE ✓
SW-8 As a user I want to add a delivery request so th...	Amal	lina	Medium	DONE ✓
SW-9 As a user, I want to view my request details an...	Amal	lina	Medium	DONE ✓
SW-10 As a user, I want to search for my request usin...	Amal	lina	Medium	DONE ✓
SW-11 As a user I want to edit my delivery request be...	lina	lina	Medium	DONE ✓
SW-12 As a user, I want to delete my delivery request...	Amal	lina	Medium	DONE ✓
SW-13 As an admin, I want to add driver accounts so ...	Albatool	lina	Medium	DONE ✓
SW-14 As an admin, I want to delete a driver account ...	Albatool	lina	Medium	DONE ✓

Spaces

2026-IT320--section1910-G4



Summary Timeline Backlog Board Calendar List Forms Development Code More 3 +

Search work Filter Group

Saved filters

Work	Assignee	Reporter	Priority	Status
SW-9 As a user, I want to view my request details an...	Amal	lina	Medium	DONE
SW-10 As a user, I want to search for my request usin...	Amal	lina	Medium	DONE
SW-11 As a user I want to edit my delivery request be...	lina	lina	Medium	DONE
SW-12 As a user, I want to delete my delivery request...	Amal	lina	Medium	DONE
SW-13 As an admin, I want to add driver accounts so ...	Albatool	lina	Medium	DONE
SW-14 As an admin, I want to delete a driver account ...	Albatool	lina	Medium	DONE
SW-15 As a user, I want to rate the delivery service af...	lina	lina	Medium	DONE
SW-16 As a user, I want to submit a written review aft...	lina	lina	Medium	DONE
SW-17 As an admin, I want to view a dashboard that s...	Albatool	lina	Medium	DONE
SW-18 As a user, I want my delivery request to includ...	Shahd aldwhan	lina	Medium	DONE
SW-20 As a user, I want the system pages to load qui...	lina	lina	Medium	DONE
SW-21 As a user, I want the system to be available w...	Amal	lina	Medium	DONE

8.1 Appendix D: User Acceptance Testing

This appendix includes the questionnaire questions used for User Acceptance Testing (UAT), the link to the collected responses, and the results graphs. These elements provide a clear overview of the participants feedback

Questionnaire Questions:

1. The steps to create a new account were clear.
2. Logging in to the system worked as expected.
3. The process of creating a delivery request was clear.
4. The list of requests was displayed clearly.
5. The request details page was understandable.
6. The search feature helped in finding a request.
7. The process of editing a request was clear.
8. The process of deleting a request before delivery was clear.
9. Submitting a rating and review after delivery was clear.
10. Logging out of the system worked as expected.

Responses Link:

<https://docs.google.com/spreadsheets/d/1ZMDLYkJe5FXegemTykJcw4CoPmgrM9YLBiaIIsuI9s/edit?usp=drivesdk>

Results Graphs:

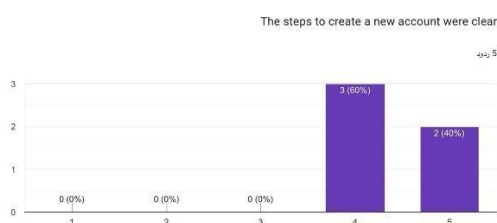


Figure 37: question 1

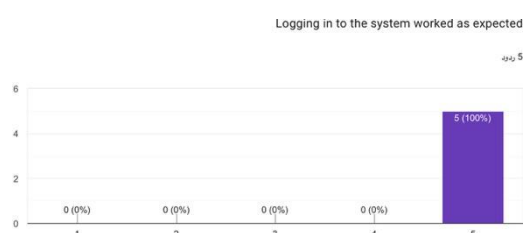


Figure 38: question 2

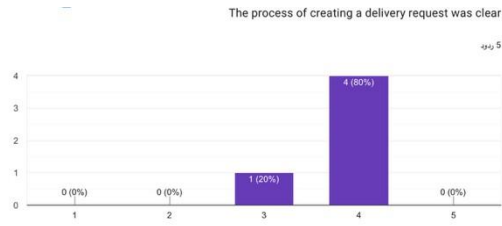


Figure 39: question 3

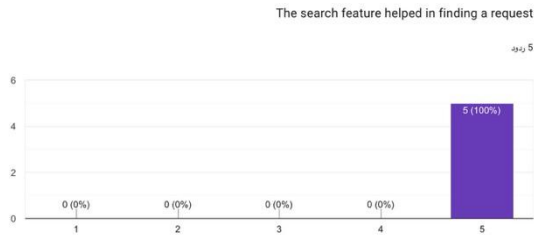


Figure 40: question 4

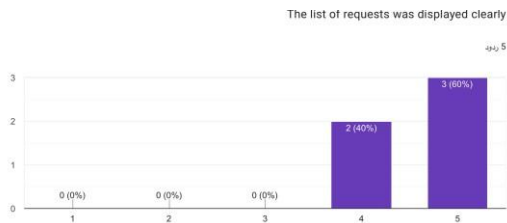


Figure 41: question 5

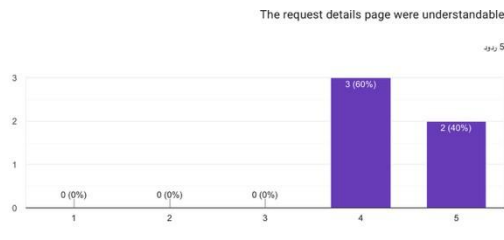


Figure 42: question 6

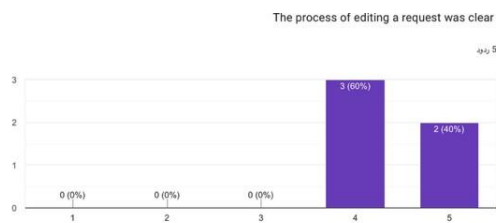


Figure 43: question 7

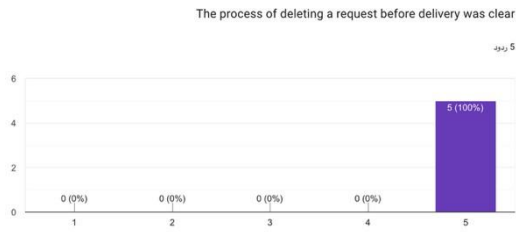


Figure 44: question 8

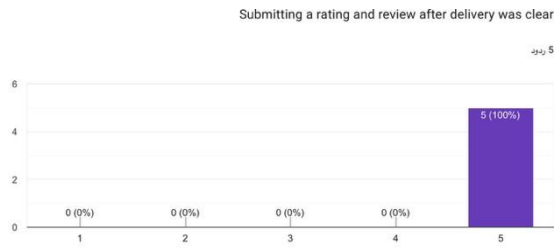


Figure 45: question 9

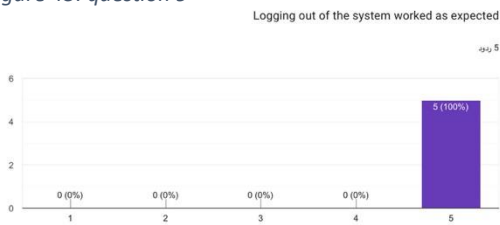


Figure 46: question 10